

RAS: Retrieval-And-Structuring for Knowledge-Intensive LLM Generation

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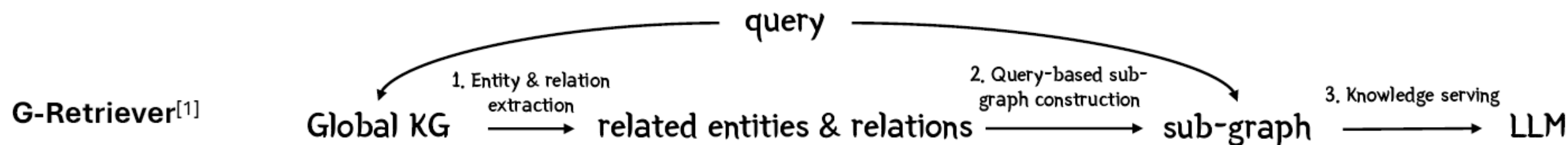
Overview

- Related Works & Problems
- Methodology
- Data Preparation & Model Training
- Experiments
- Conclusion & Future Work

Related Works & Problems

Related Works & Problems

I. Graph-based RAG

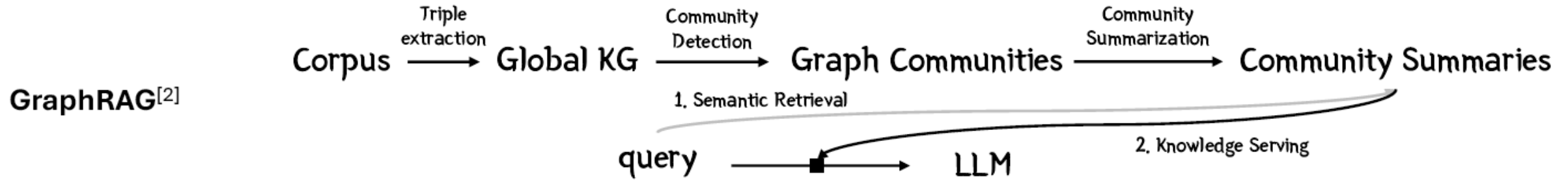


Problem of G-Retriever:

- (1) The quality of knowledge highly depends on the initial entity/relation extraction, which is usually bad as it is based on the semantic matching.
 - the most helpful context doesn't necessarily have the entities or relations that are “most relevant” to the query itself.
- (2) No knowledge refinement process.
- (3) Agnostic on what LLM needs.

Related Works & Problems

I. Graph-based RAG



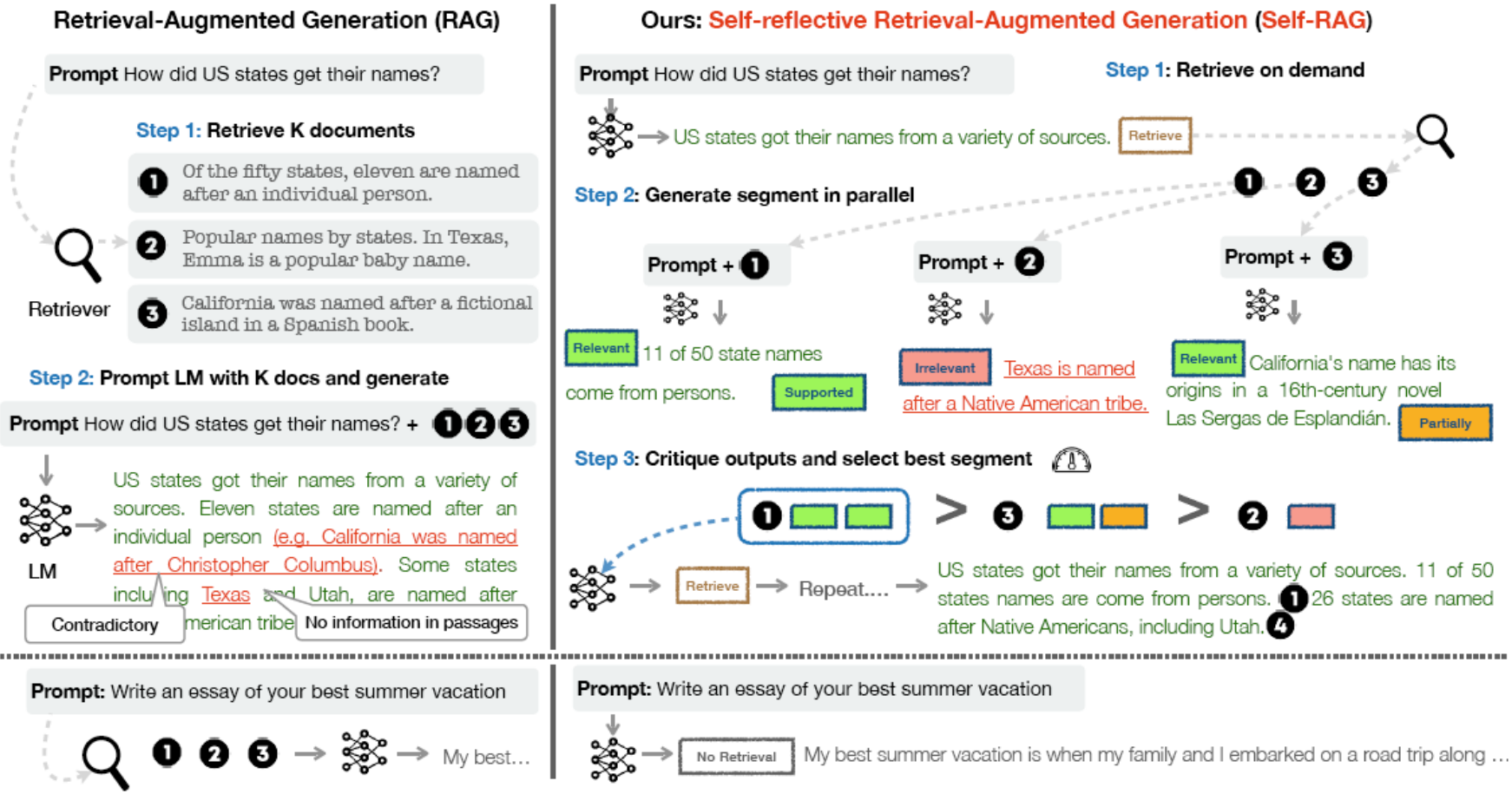
Problem of GraphRAG:

- (1) The quality of knowledge highly depends on the graph communities detected, which have unreliable quality as the communities are detected solely based on the structural information.
- (2) No knowledge refinement process.
- (3) Agnostic on what LLM needs.

Related Works

II. Self-Reflective RAG

SelfRAG[3]

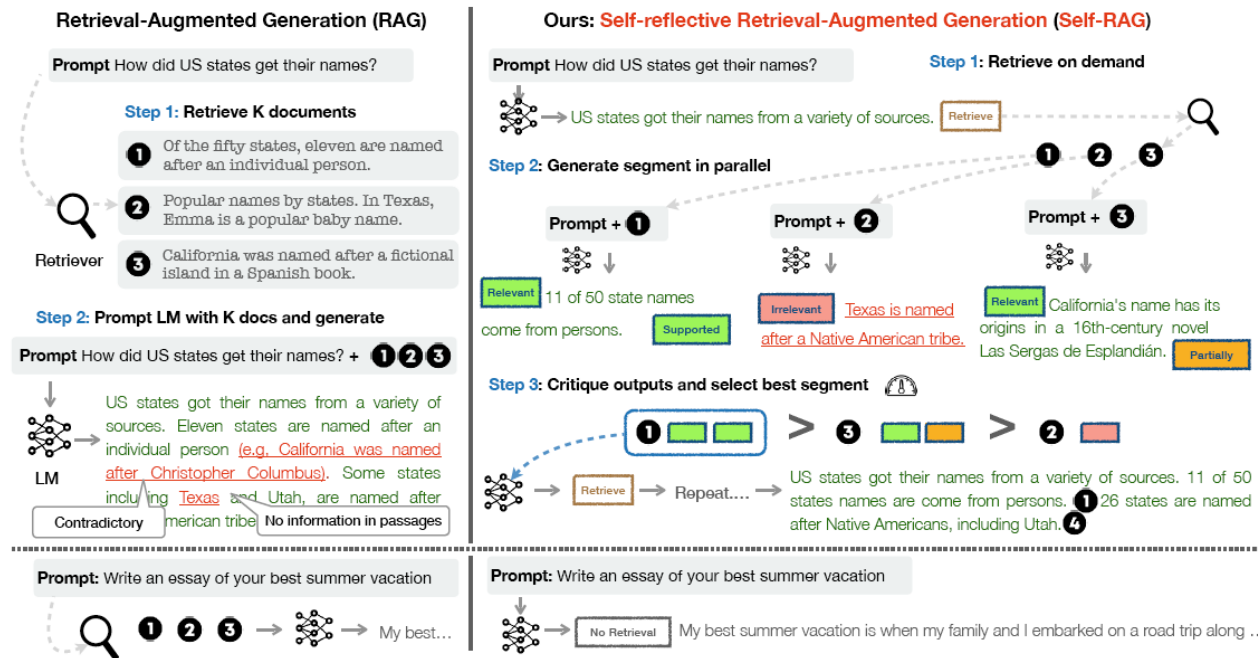


[3] Asai, Akari, et al. "Self-rag: Learning to retrieve, generate, and critique through self-reflection." in ICLR 2024.

Related Works

II. Self-Reflective RAG

SelfRAG^[3]



Pros:

- (1) Retrieve context on LLM's demand
- (2) Iterative context enrichment

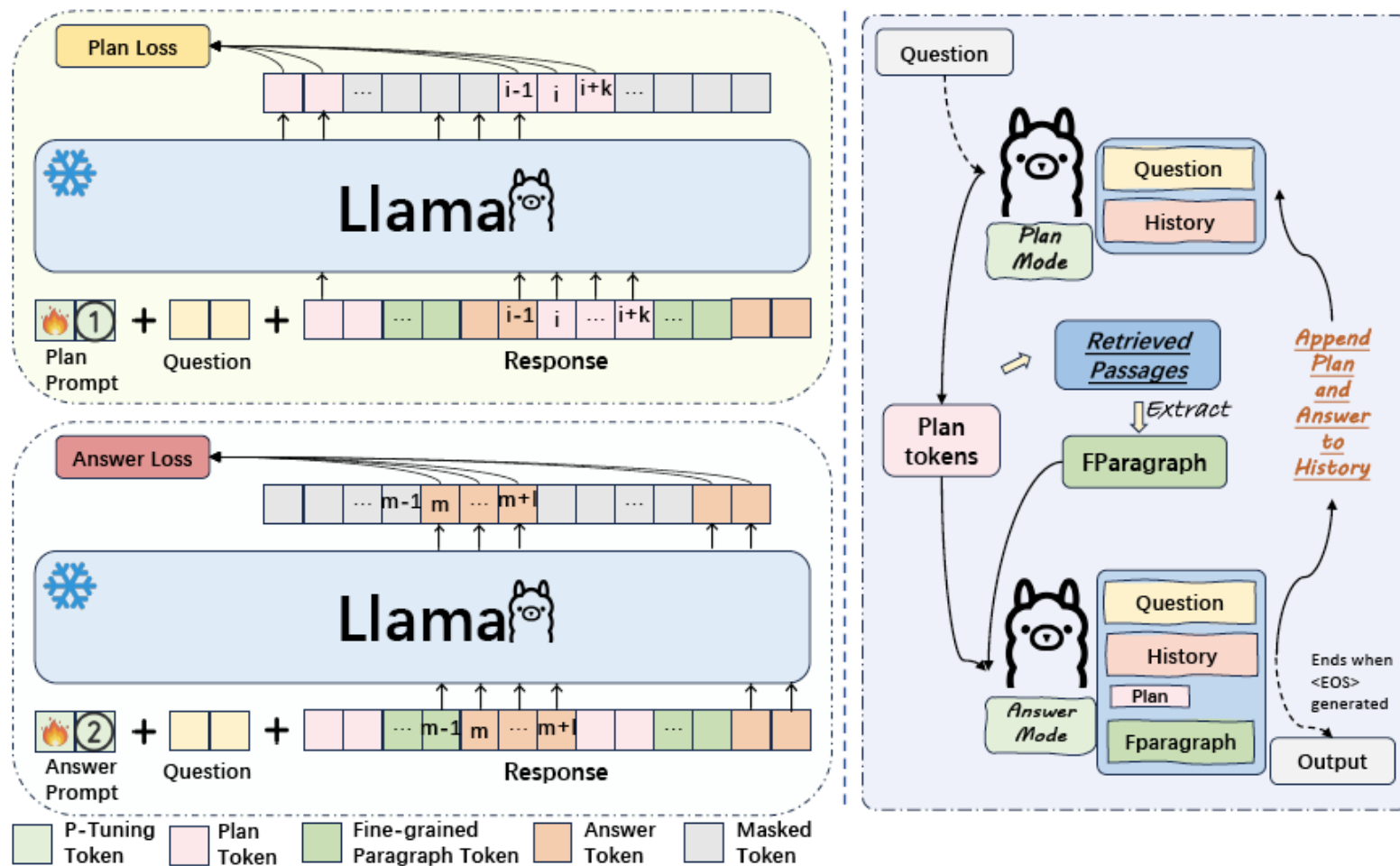
Cons:

- (1) Searching scope of retrieval is too big, and purely based on semantics
- (2) Retrieved knowledge is not managed well (no structural info)
- (3) No knowledge refinement

Related Works

II. Self-Reflective RAG

RPG[4]



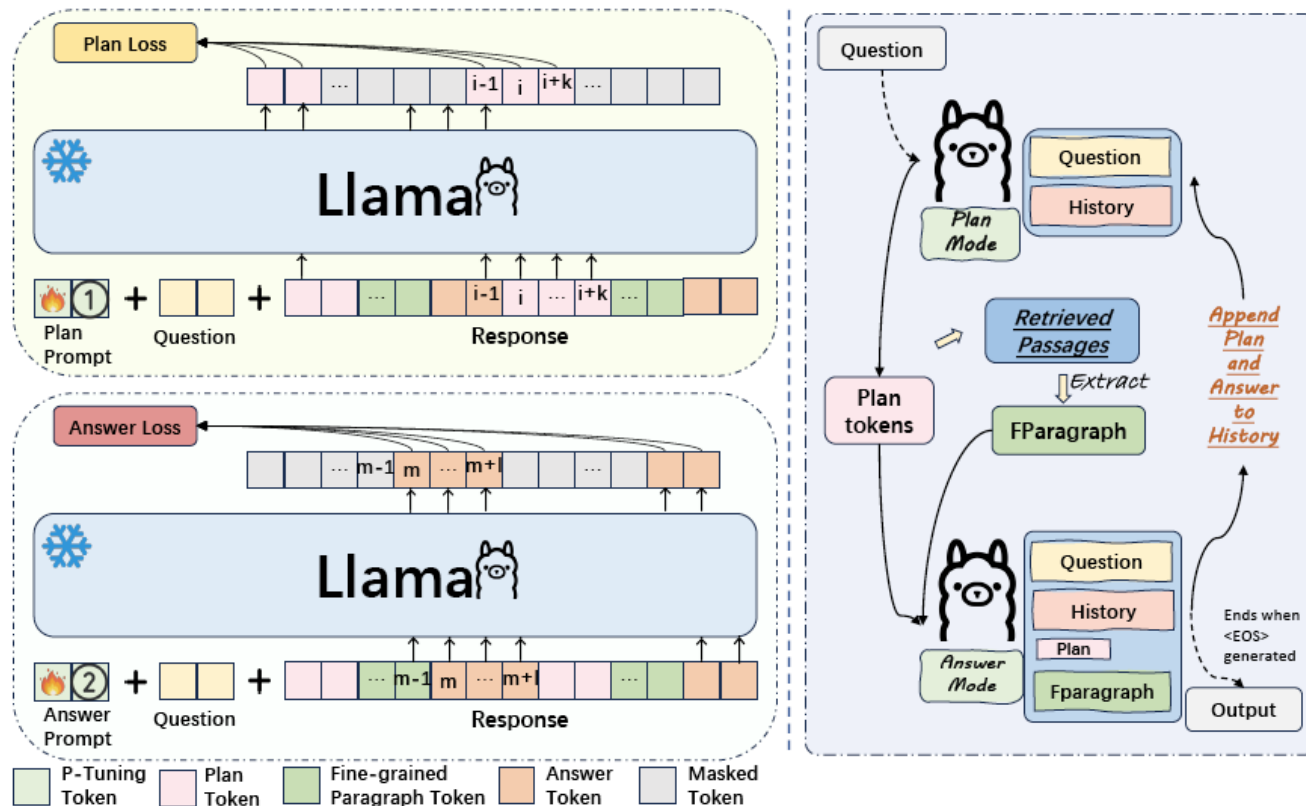
Most novel module of RPG:
Extract **FParagraph**

“Explicitly relevant part of
the retrieved context”

Related Works

II. Self-Reflective RAG

RPG^[4]



Pros:

- (1) Refine the retrieved context by extracting its “FParagraph”
- (2) Plan before generate

Cons:

- (1) Searching scope of retrieval is too big, and purely based on semantics
- (2) No structural knowledge

Related Works

What We Want

Graph-based RAG

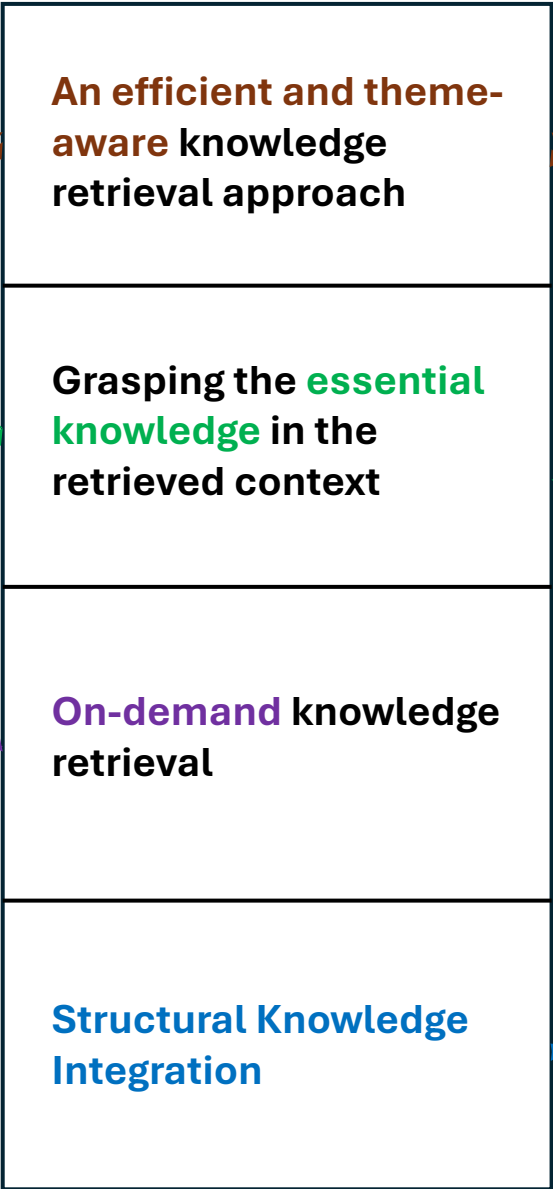
Cons of G-Retriever:

- (1) Poor knowledge retrieval
- (2) No knowledge refinement process.
- (3) Agnostic on what LLM needs.

Cons of GraphRAG:

- (1) Arbitrary community-based knowledge indexing
- (2) No knowledge refinement process.
- (3) Agnostic on what LLM needs.

Building corpus-wise KG is very costly →
On-the-fly structuring is preferred



Self-Reflective RAG

Cons of SelfRAG

- (1) Searching scope of retrieval is too big, and purely based on semantics
- (2) Retrieved knowledge is not managed well (no structural info)
- (3) The retrieved context is lengthy; No knowledge refinement process

Cons of RPG

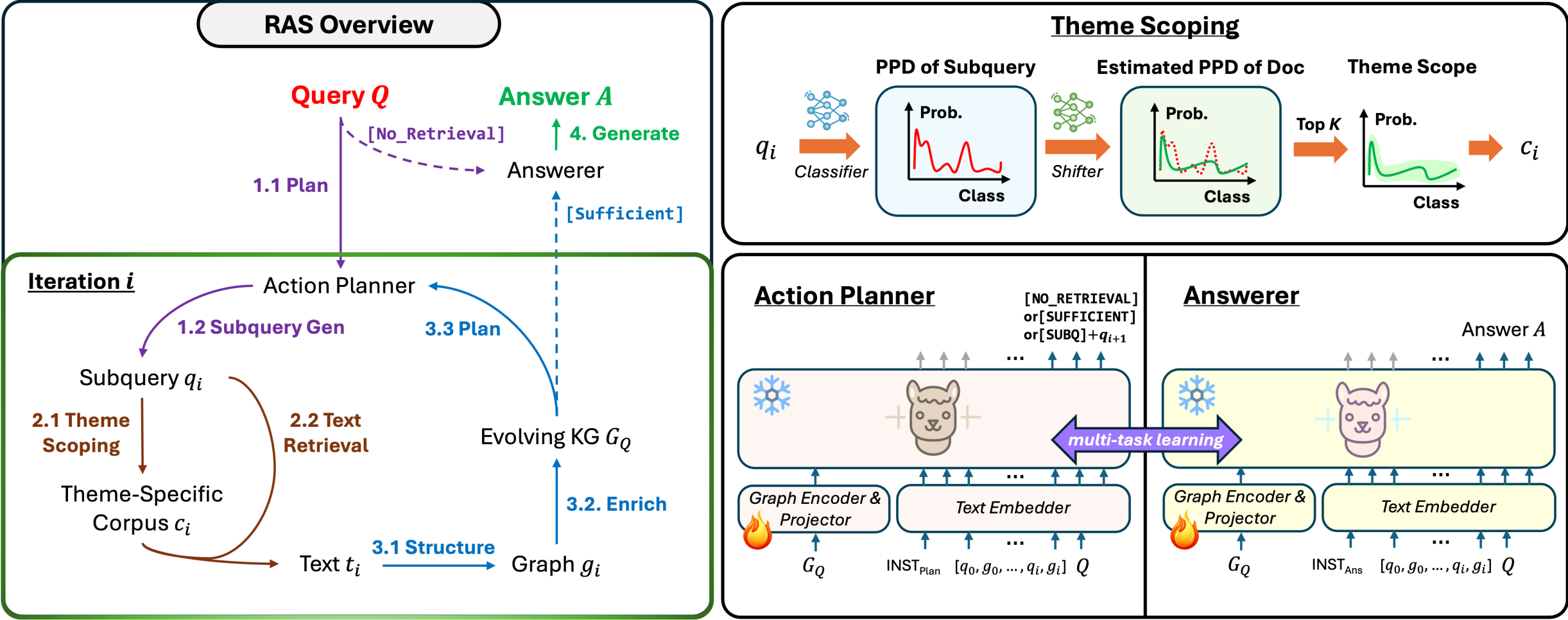
- (1) Searching scope of retrieval is too big, and purely based on semantics
- (2) No structural knowledge

Both SelfRAG and RPG cannot be directly applied to closed-source LLMs

Methodology

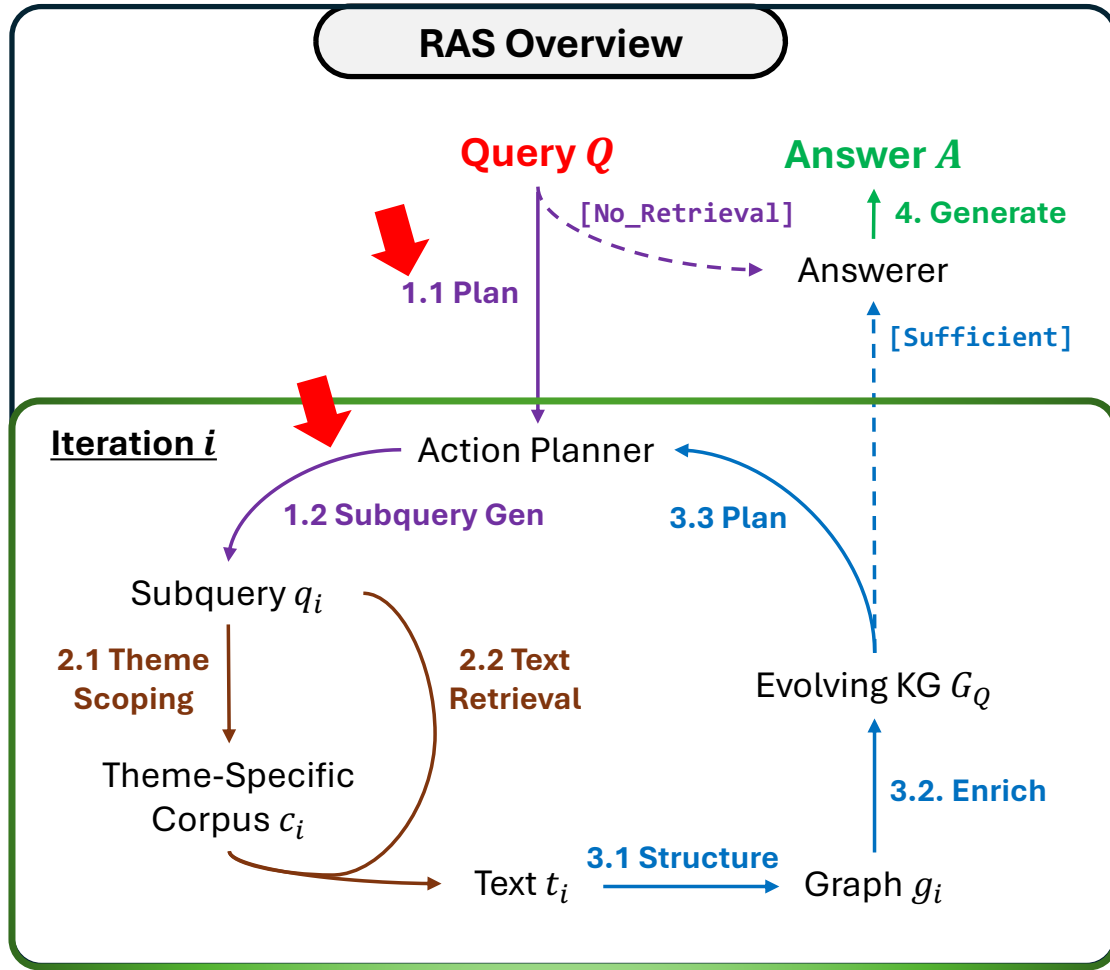
Methodology

RAS – Retrieval-And-Structuring for Knowledge-Intensive LLM Generation



Methodology

RAS Framework



Step 1:

When a query Q comes in, we ask the Action Planner $\mathcal{M}_{Plan}$ to output either

- **[NO_RETRIEVAL]** if the query can be answered w/o any retrieval, or
- **“[SUBQ] + The first subquery q_0 ”** if the query needs more context to be answered

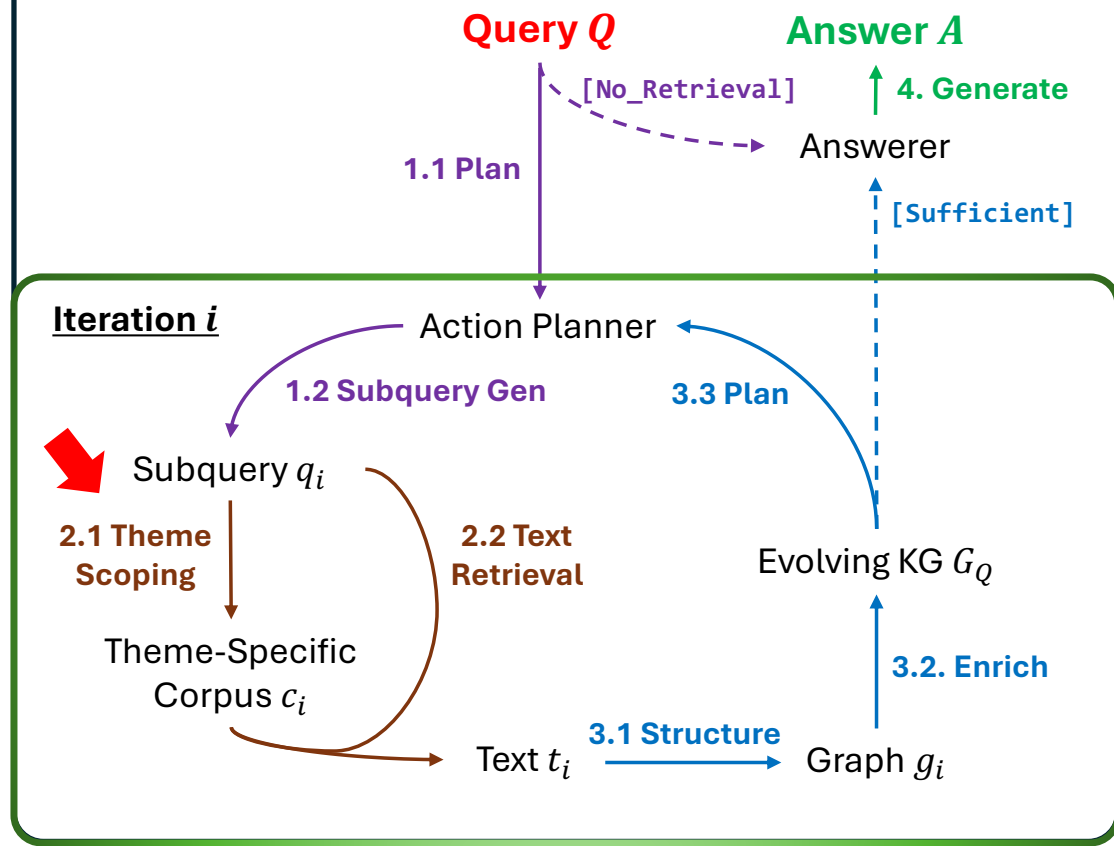
If **[NO_RETRIEVAL]**, we skip intermediate steps and jump to Step 4 for LLM generation.

If a subquery is generated, we move forward to Step 2.

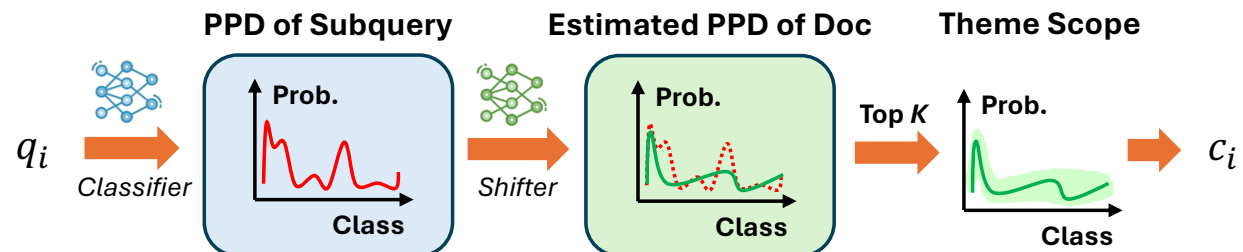
Methodology

RAS Framework

RAS Overview



Theme Scoping



Step 2:

When a subquery q_i comes in, we conduct:

(2.1 Theme Scoping)

Use a multilabel document classifier f_{cls} to output the posteriori (class) probability distribution (PPD) $PPD(q_i)$ of q_i

$$PPD(q_i) \leftarrow f_{cls}(q_i)$$

Shift $PPD(q_i)$ to the estimated PPD of document $PPD(d_i)$, with a distribution shifter f_{shift}

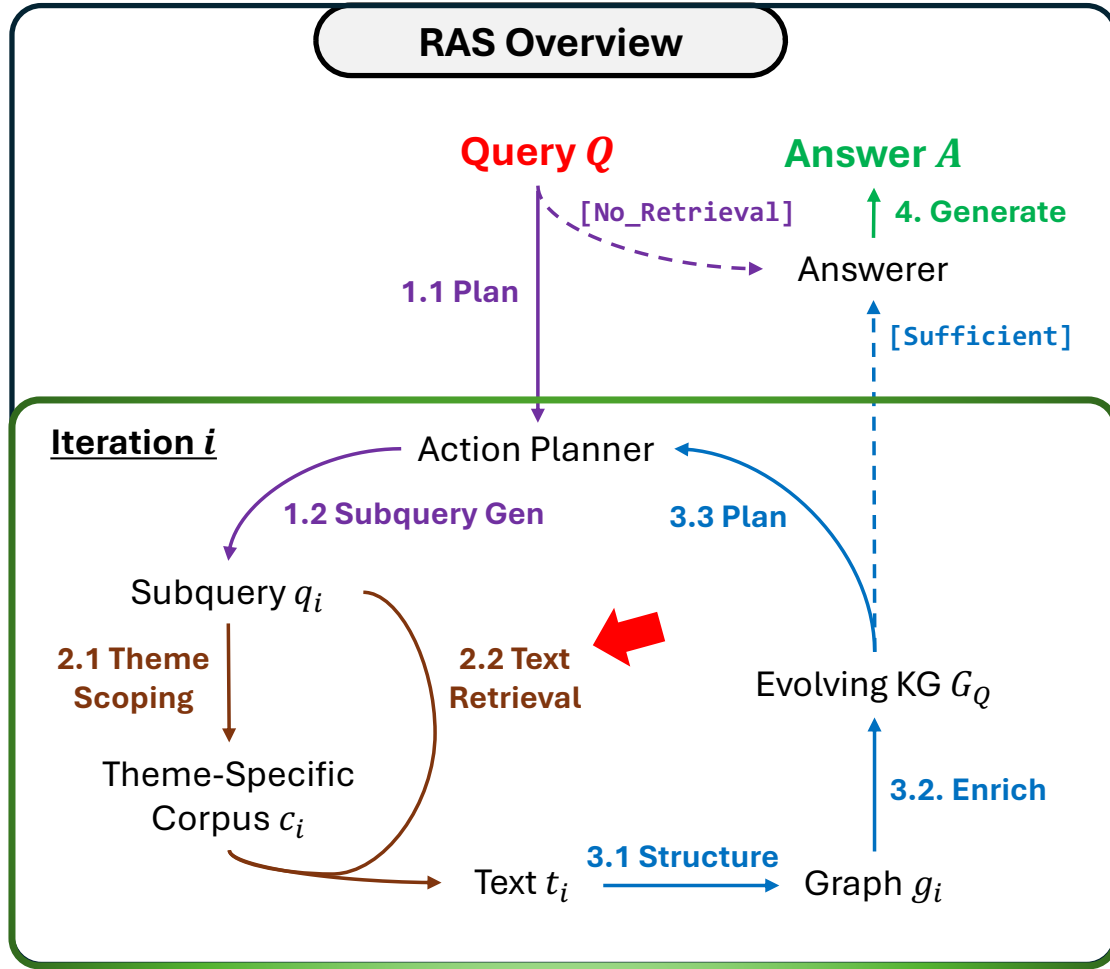
$$PPD(d_i) \leftarrow f_{shift}(PPD(q_i))$$

With $PPD(d_i)$, we retrieve top K documents from global corpus C with its theme index C_{PPD} (created by f_{cls}), as theme-specific corpus c_i

$$c_i \leftarrow \text{ThemeSim}(PPD(d_i), C_{PPD}, K)$$

Methodology

RAS Framework



Step 2:

With a subquery q_i and its theme-specific corpus c_i :

(2.2 Text Retrieval)

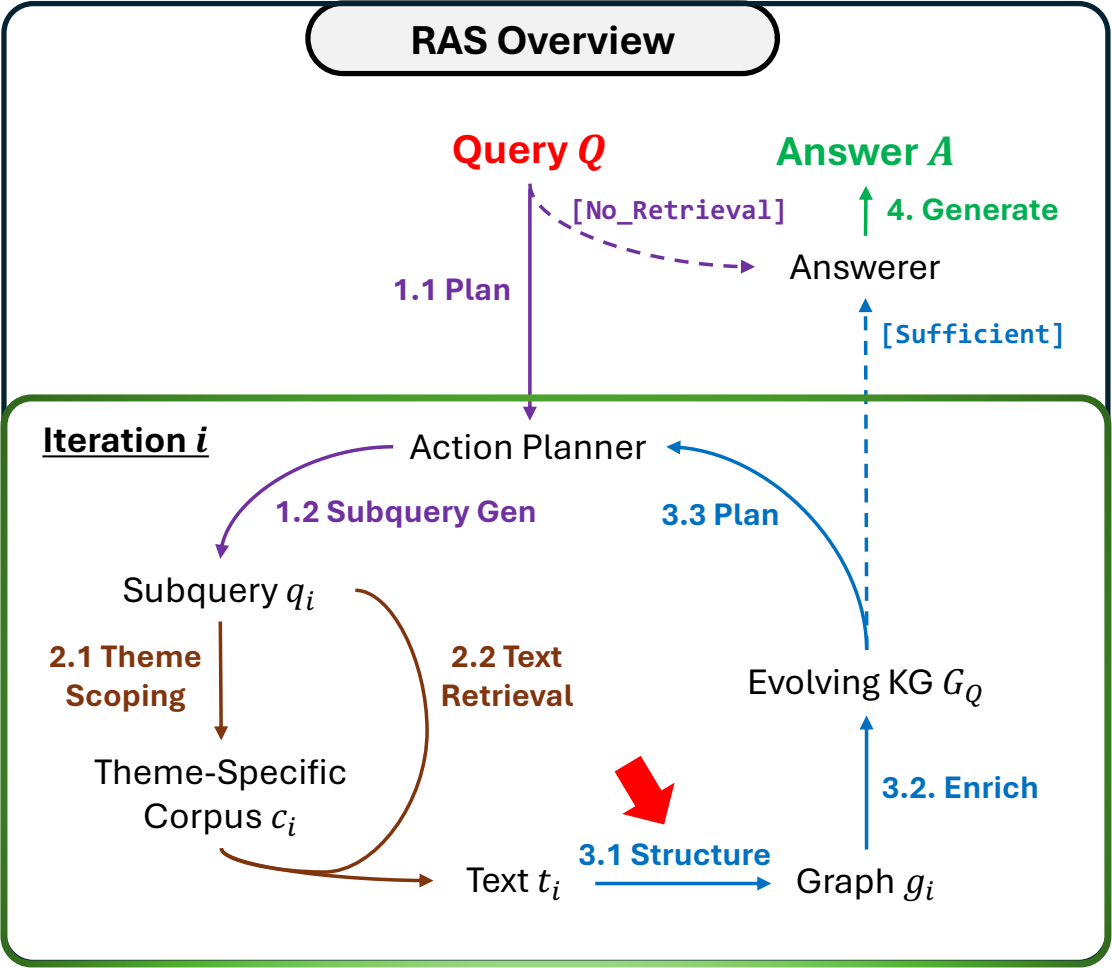
Using a dense retriever f_{den} we retrieve top k documents from c_i with its dense index (created by f_{den}), as t_i

$$t_i \leftarrow \text{DenseSim}(f_{den}(q_i), f_{den}(c_i), k)$$

Note: in implementation, the global corpus $\mathcal{C}$ is pre-indexed and $f_{dense}(c_i)$ is just a process selecting the dense embeddings of the documents in c_i , which is cost-efficient and speedy.

Methodology

RAS Framework



Step 3:

(3.1 Structure)

With the retrieved text t_i , we use a text-to-triples model f_{t2t} to extract essential graph(s) from it.

$$g_i \leftarrow f_{t2t}(t_i)$$

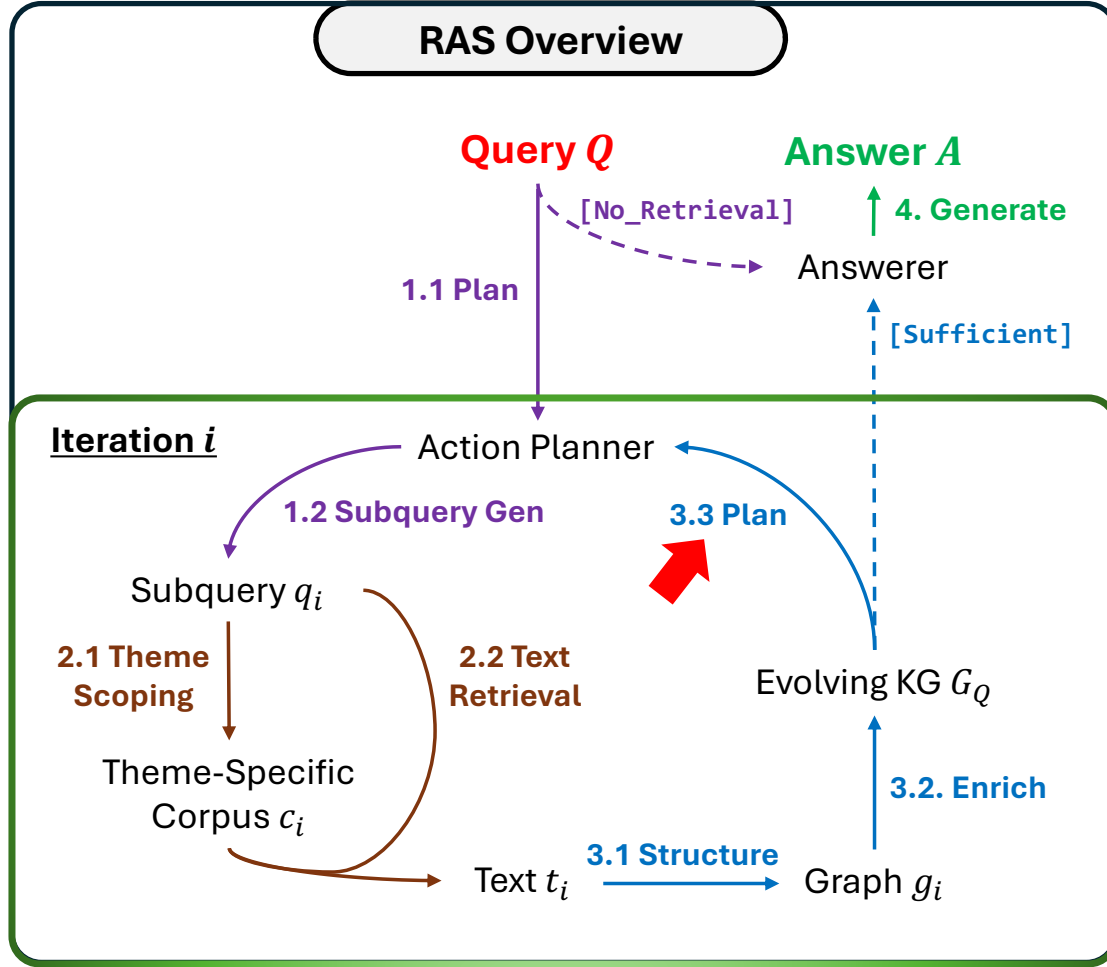
(3.2 Enrich)

At each iteration, we merge the graph g_i (specific to the subquery q_i) to the Evolving KG G_Q (specific to the main query Q)

$$G_Q = [g_0, g_1, \dots, g_i] \text{ at iteration } i$$

Methodology

RAS Framework



Step 3:

(3.3 Plan)

With

- Evolving KG G_Q ,
- Subqueries $[q_0, q_1, \dots, q_i]$,
- Triples $[g_0, g_1, \dots, g_i]$,
- Query Q

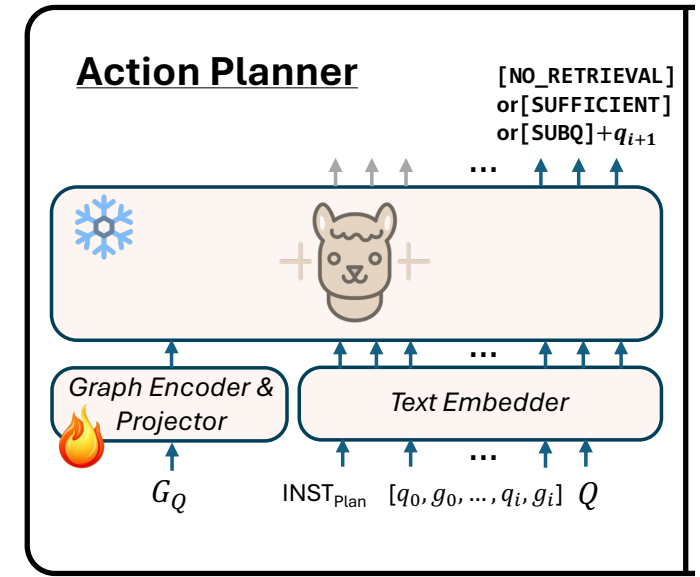
we ask the Action Planner the next step:

$$p_{i+1} \leftarrow \mathcal{M}_{Plan}([f_{GNN}(G_Q); (q_0, g_0), \dots, (q_i, g_i); Q])$$

where f_{GNN} is a GNN graph encoder, $\text{text}(g_i)$ is the triple list of g_i in string format.

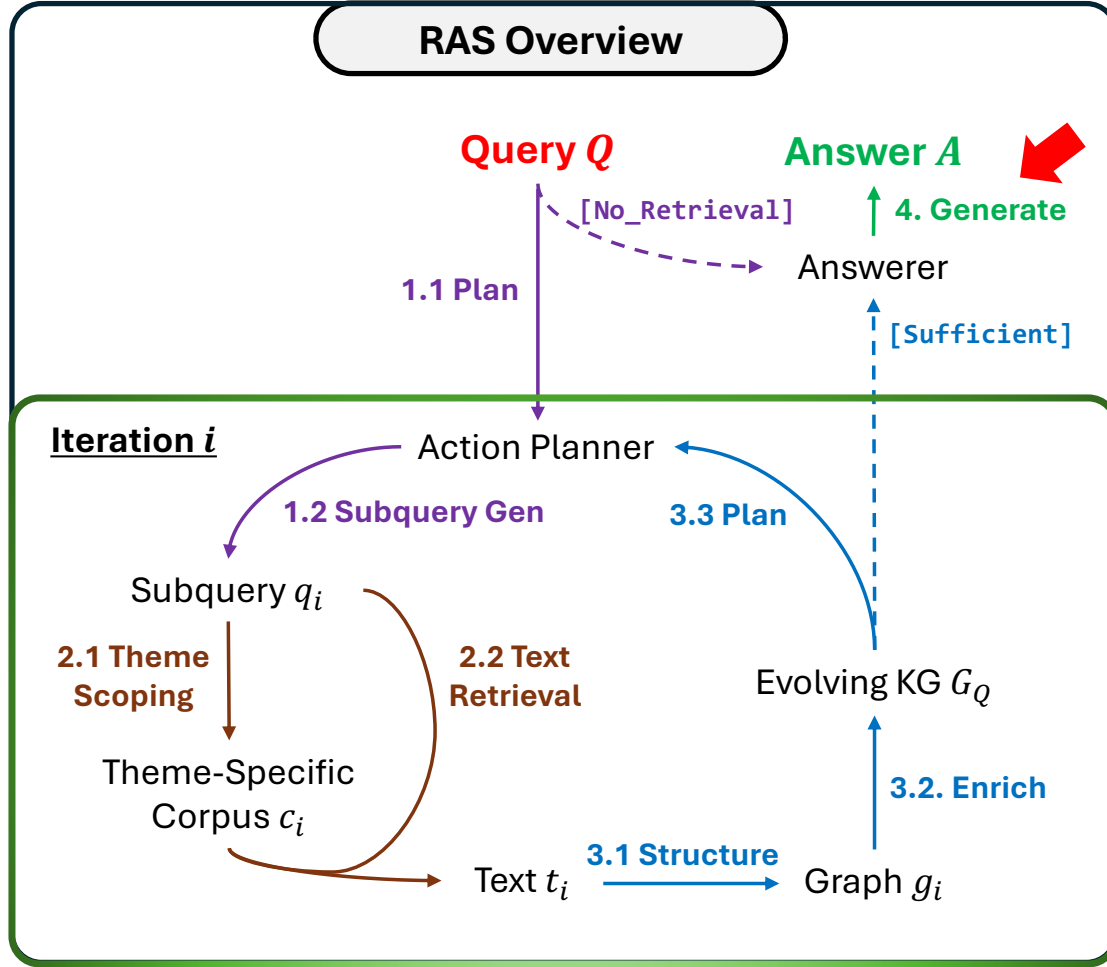
The plan p_{i+1} can either be

- **[Sufficient]** if the information is enough to answer the query Q
- **“[SUBQ] + q_{i+1} ”** if the information is not enough.



Methodology

RAS Framework



Step 4:

With

- Evolving KG G_Q ,
- Subqueries $[q_0, q_1, \dots, q_i]$,
- Triples $[g_0, g_1, \dots, g_i]$,
- Query Q

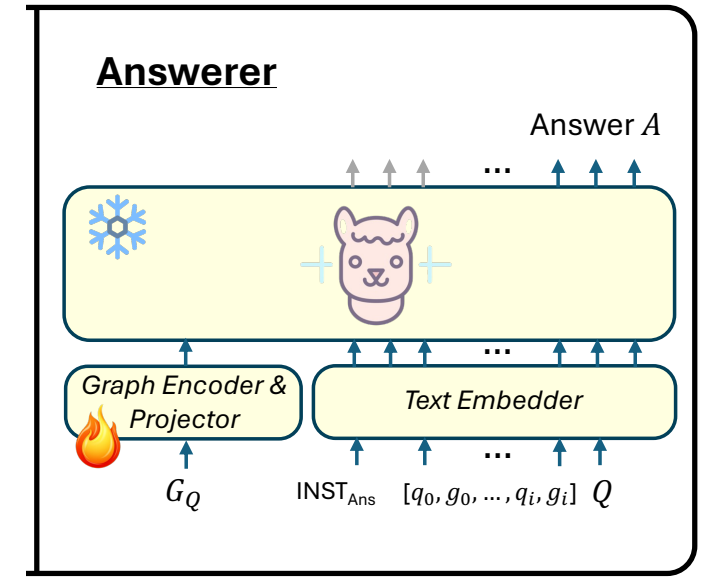
When (1) no retrieval is needed, or (2) the information is sufficient, we ask the Answerer $\mathcal{M}_{Ans}$ to generate the answer:

If [NO_RETRIEVAL]:

$$A \leftarrow \mathcal{M}_{Ans}([None; None; Q])$$

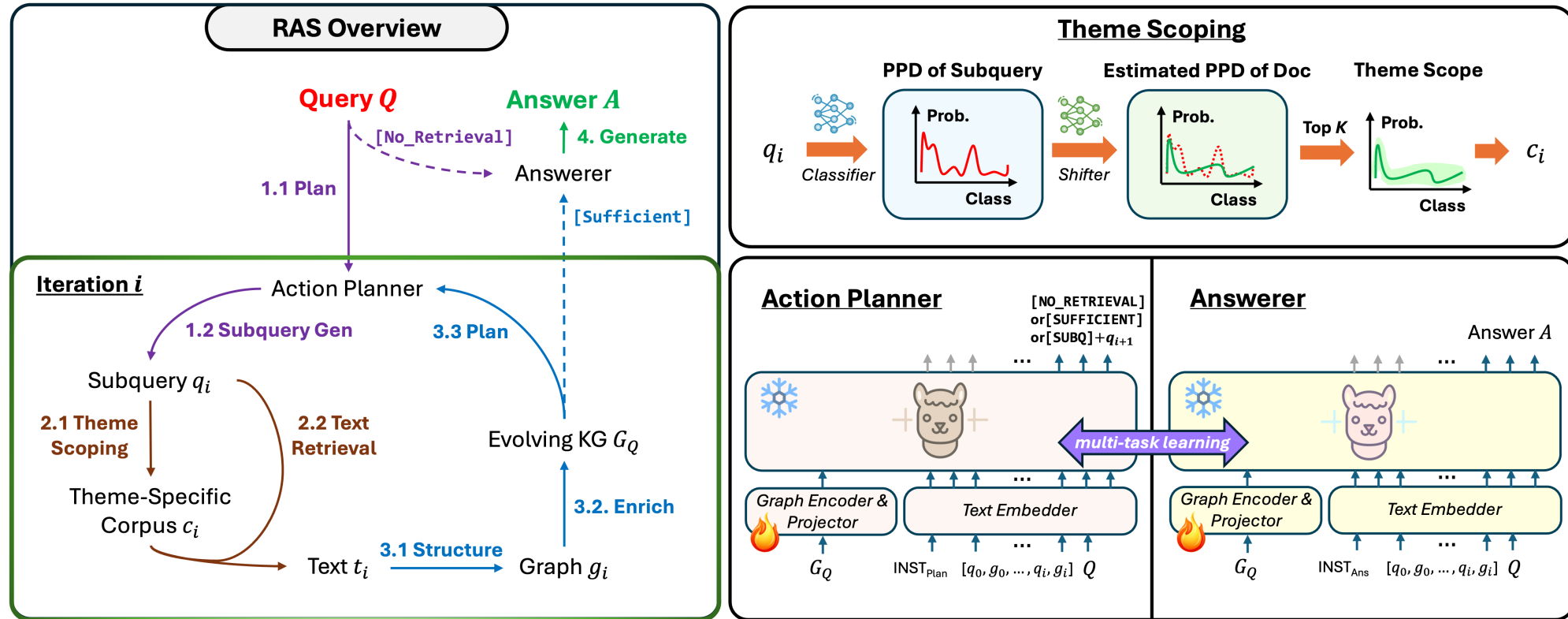
If [Sufficient]:

$$A \leftarrow \mathcal{M}_{Ans}([f_{GNN}(G_Q); (q_0, g_0), \dots, (q_i, g_i); Q])$$



Methodology

RAS – Retrieval-And-Structuring for Knowledge-Intensive LLM Generation



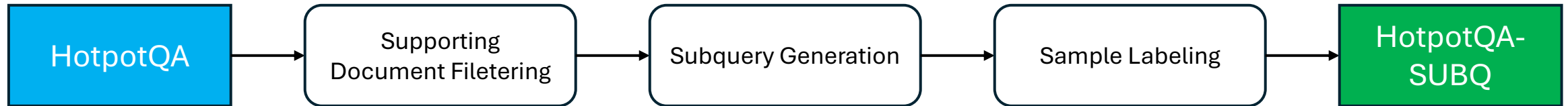
There is a need to construct a dataset, from which RAS can learn action planning and answering capabilities with structured data.

Data Preparation & Model Training

Data Preparation & Model Training

Data Preparation – HotpotQA-SUBQ

We construct **HotpotQA-SUBQ**, a dataset containing multi-hop QA samples where each supporting document has its corresponding subquery.

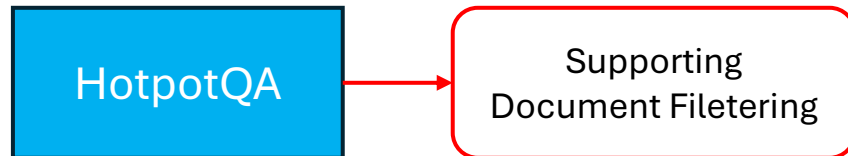


Data Preparation & Model Training

Data Preparation – HotpotQA-SUBQ

HotpotQA is not clean, some supporting documents are unnecessary or even irrelevant to answer the questions.

→ we need to filter out those docs



Identify which documents are HELPFUL to answer the question. Output only the document numbers separated by commas.

Examples:

Example 1 (Some documents are not helpful):

Question: What nationality was James Henry Miller's wife?

Supporting docs:

1. Margaret "Peggy" Seeger (born June 17, 1935) is an American folksinger. She is also well known in Britain, where she has lived for more than 30 years, and was married to the singer and songwriter Ewan MacColl until his death in 1989.
2. Seeger's father was Charles Seeger (1886–1979), an important folklorist and musicologist; her mother was Seeger's second wife, Ruth Porter Crawford.
3. James Henry Miller, better known by his stage name Ewan MacColl, was an English folk singer and songwriter.

Output: 1,3

Explanation: Only docs 1 and 3 are helpful - doc 1 shows Peggy Seeger (who is American) was married to Ewan MacColl, and doc 3 confirms Ewan MacColl is James Henry Miller. Doc 2 about Seeger's parents is not helpful.

Example 2 (All documents are helpful):

Question: The Oberoi family is part of a hotel company that has a head office in what city?

Supporting docs:

1. The Oberoi family is an Indian family that is famous for its involvement in hotels, namely through The Oberoi Group.
2. The Oberoi Group is a hotel company with its head office in Delhi.

Output: 1,2

Explanation: Both docs are helpful - doc 1 links the Oberoi family to The Oberoi Group, and doc 2 provides the head office location.

Question: [question]

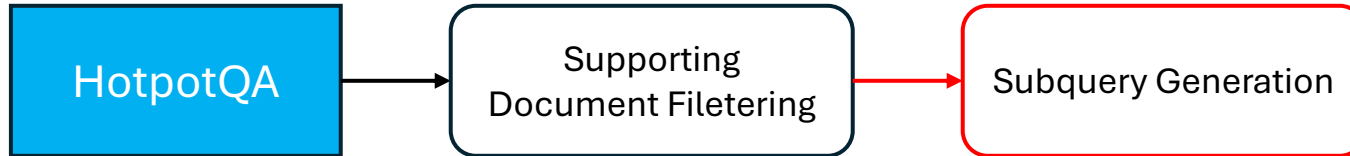
Supporting docs:

[enumerated documents]

Output only the helpful document numbers separated by commas:

Data Preparation & Model Training

Data Preparation – HotpotQA-SUBQ



For each supporting document, we generate a subquery that helps to retrieve it for progressive information gathering to answer the main question.

Given this main question and a supporting document, generate a simple sub-query (a question) that will help retrieve information from the document to answer the main question.

Main Question: **[main_question]**

Current Document ([topic]):

[document_content]

[If previous queries exist:]

Previously generated sub-queries:

- **[sub_query_1]**

- **[sub_query_2]**

...

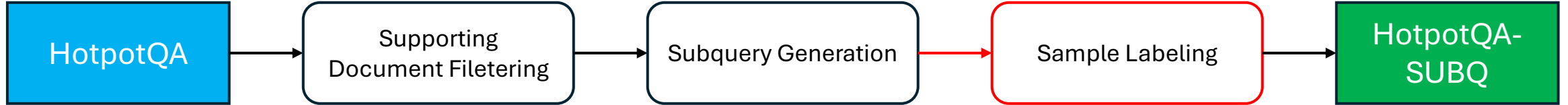
Write ONE clear and specific question that:

1. Can be answered using ONLY this document
2. Helps retrieve information needed for the main question
3. Is direct and focused on key information from this document

Write only the question, without any explanations or formatting.

Data Preparation & Model Training

Data Preparation – HotpotQA-SUBQ



After gathering processed HotpotQA, we label the data as follows:

[NO_RETRIEVAL] :

- We use LLaMA-2-7B itself to answer the questions in HotpotQA, and label those correct answered ones as [NO_RETRIEVAL]

[SUBQ] + q_{i+1} :

- If q_i is not the subquery for the last document

[SUFFICIENT] :

- If q_i is the subquery for the last document

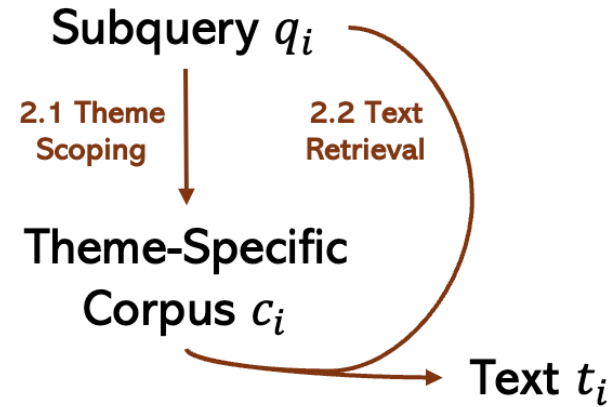
On the way of labeling, we transform each document into a triple list using our trained model f_{t2t}

		Planner Data	Answerer Data
# Queries		129,902	78,164 (w/ 3,897 ASQA & 2,147 Arc-Easy)
# Input Tokens	(Mean, Median, Max)	(338, 301, 1,910)	(475, 466, 2,214)
# Output Tokens	(Mean, Median, Max)	(13, 12, 62)	(13, 4, 2,332)
# Subqueries	(Min, Mean, Max)	(0, 0.8, 5)	(0, 1.2, 6)
# [SUFFICIENT]		45,722	–
# [SUBQ]		71,676	–
# [NO_RETRIEVAL]		12,504	–
# [SUBQ] Tokens	(Min, Mean, Max)	(6, 18.3, 62)	–
# Nodes	(Mean, Median, Max)	(11.6, 11, 56)	(11.7, 11, 84)
# Edges	(Mean, Median, Max)	(21.2, 20, 116)	(21.3, 20, 160)

Table 3. Statistics of our constructed HotpotQA-SUBQ dataset.

Data Preparation & Model Training

Models Training for Theme-Scoping and Text Retrieval



$$\text{PPD}(q_i) \leftarrow f_{cls}(q_i)$$

$$\text{PPD}(d_i) \leftarrow f_{shift}(\text{PPD}(q_i))$$

$$c_i \leftarrow \text{ThemeSim}(\text{PPD}(d_i), C_{PPD}, K)$$

$$t_i \leftarrow \text{DenseSim}(f_{den}(q_i), f_{den}(c_i), k)$$

Multilabel Document Classifier f_{cls} :

Data: DBPedia-298

Distribution Shifter f_{shift} :

Base model: MLP

Data: HotpotQA-SUBQ question-document pairs

Dense Retriever f_{den} :

Contriever-MSMARCO (align with previous works)

Data Preparation & Model Training

Model Training for Text-to-Triples Structuring

Text t_i $\xrightarrow{\text{3.1 Structure}}$ Graph g_i

$$g_i \leftarrow f_{t2t}(t_i)$$

(Models are released on HF)

[pat-jj/text2triple-flan-t5](#)

[pat-jj/text2graph-llama-3.2-3b](#)

Text-to-Triples Transformer f_{t2t} :

Base Model: Flan-T5-Large

Data: WikiOfGraph (5.85M high quality text-triple pairs)

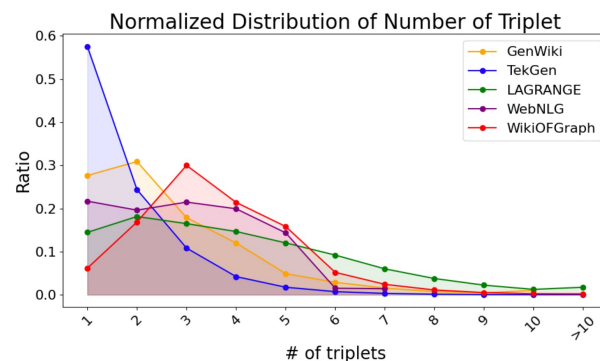


Figure 3: Normalized distribution of the number of triplets in each dataset.

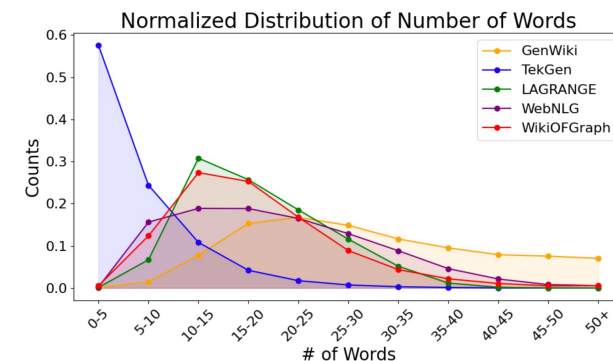
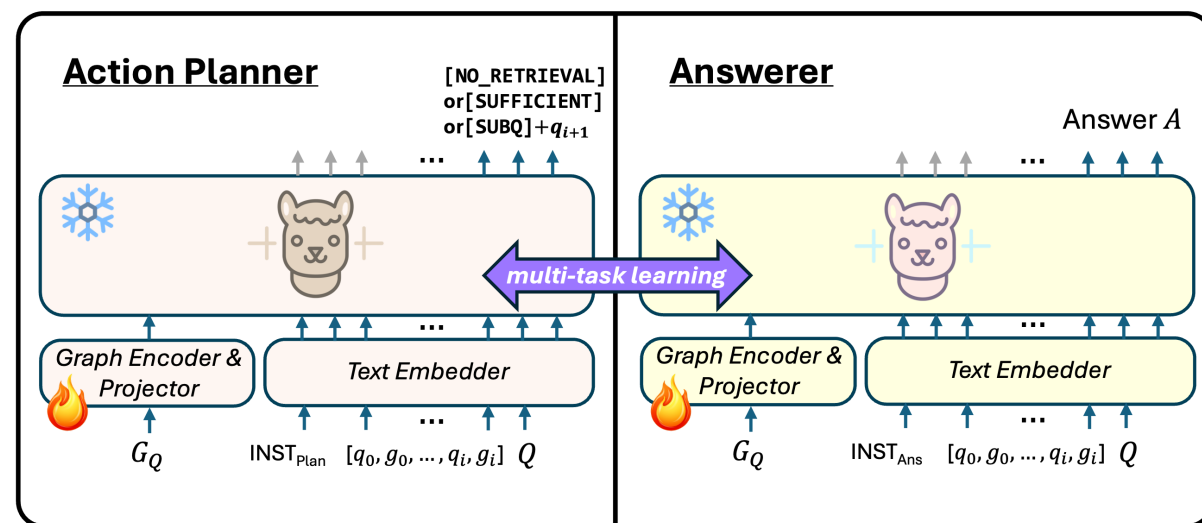


Figure 4: Normalized distribution of the number of words in each dataset.

Data Preparation & Model Training

Model Training for Action Planning and Answering with Structured Data



		Planner Data	Answerer Data
# Queries		129,902	78,164 (w/ 3,897 ASQA & 2,147 Arc-Easy)
# Input Tokens	(Mean, Median, Max)	(338, 301, 1,910)	(475, 466, 2,214)
# Output Tokens	(Mean, Median, Max)	(13, 12, 62)	(13, 4, 2,332)
# Subqueries	(Min, Mean, Max)	(0, 0.8, 5)	(0, 1.2, 6)
# [SUFFICIENT]		45,722	–
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# [SUBQ] Tokens	(Min, Mean, Max)	(6, 18.3, 62)	–
# Nodes	(Mean, Median, Max)	(11.6, 11, 56)	(11.7, 11, 84)
# Edges	(Mean, Median, Max)	(21.2, 20, 116)	(21.3, 20, 160)

Table 3. Statistics of our constructed HotpotQA-SUBQ dataset.

$$p_0 \leftarrow \mathcal{M}_{Plan}([None; None; Q])$$

$$A \leftarrow \mathcal{M}_{Ans}([None; None; Q])$$

$$p_{i+1} \leftarrow \mathcal{M}_{Plan}([f_{GNN}(G_Q); \\ (q_0, g_0), \dots, (q_i, g_i); Q])$$

$$A \leftarrow \mathcal{M}_{Ans}([f_{GNN}(G_Q); \\ (q_0, g_0), \dots, (q_i, g_i); Q])$$

Experiments

Experiments

Experimental Setting

We test our RAS Framework in the zero-shot inference setting, with both the *closed-source* and the *open-source* base LLMs

- **Closed-source:** We use Claude 3.5 Sonnet as the LLM, where examples are included in the instruction prompts for action planning and answering.
- **Open-source:** We use LLaMA-2-7B/LLaMA-3-8B as the LLMs, which are fine-tuned in the GraphLLM framework with our constructed HotpotQA-SUBQ dataset.

Following SelfRAG (Asai et al., ICLR 2024), we use Wikipedia 2018 dump as the knowledge source by default, while using Wikipedia 2020 for PopQA.

Experiments

Main Results

- (1) With open-source LLM (LLaMA-2-7B), RAS achieved the best performance across 8/9 datasets;
- (2) With closed-source LLM (Sonnet 3.5), RAS achieved the best performance across 4/9 datasets.
- (3) For some datasets (e.g., TQA, PubHealth), context retrieval hurts the performance of Sonnet. RAS can fix that at some degrees.
- (4) Our “role-swapping” study reveals that the upper bound of the performance is determined by Answer’s capability.

Model	Short-form			Closed-set		Long-form Generation			
	TQA (acc)	2WQA (F1)	PopQA (acc)	Pub (acc)	ARC (acc)	ASQA		ELI5	
	(rouge)	(mauve)	(rouge)	(mauve)					
<i>Baselines without retrieval</i>									
Llama2 _{7B}	30.5	18.9	14.7	34.2	21.8	15.3	19.0	18.3	32.4
Alpaca _{7B}	54.5	–	23.6	49.8	45.0	29.4	61.7	–	–
Llama2 _{13B}	38.5	20.2	14.7	29.4	29.4	12.4	16.0	18.2	41.4
Alpaca _{13B}	61.3	–	24.4	55.5	54.9	32.0	70.6	–	–
ChatGPT _{~175B}	74.3	24.8	29.3	70.1	75.3	36.2	68.8	22.8	32.6
Sonnet-3.5 _{~175B}	78.4	40.0	30.2	83.7	88.5	37.0	39.1	21.8	26.5
<i>Baselines with retrieval (#docs=5 by default)</i>									
Llama2 _{7B}	42.5	21.0	38.2	30.0	48.0	22.1	32.0	18.6	35.3
Alpaca _{7B}	64.1	–	46.7	40.2	48.0	33.3	57.9	–	–
Llama2 _{13B}	47.0	31.2	45.7	30.2	26.0	20.5	24.7	18.6	42.3
Alpaca _{13B}	66.9	–	46.1	51.1	57.6	36.7	56.6	–	–
RECOMP _{7B} (Xu et al., 2023)	–	32.4	–	–	–	36.5	76.0	–	–
SuRe _{7B} (Kim et al., 2024b)	–	20.6	–	–	–	35.8	76.2	–	–
SuRe _{GPT}	72.3	38.1	53.6	57.2	79.6	36.0	74.2	–	–
ChatGPT	65.7	32.8	50.8	54.7	75.3	39.9	79.7	20.6	57.2
Sonnet-3.5 _{#docs=1}	69.1	41.9	51.5	49.1	88.6	–	–	–	–
Sonnet-3.5 _{#docs=5}	72.5	53.7	57.3	53.9	87.1	38.8	61.6	20.2	32.3
SuRe _{Sonnet-3.5}	76.8	37.6	41.2	62.8	91.6	30.2	69.9	15.4	27.2
<i>Baselines with self-reflective retrieval</i>									
SELF-RAG _{7B} (Asai et al., 2023)	66.4	25.1	54.9	72.4	67.3	35.7	74.3	17.9	35.6
SELF-RAG _{13B}	69.3	–	55.8	74.5	73.1	37.0	71.6	–	–
RPG _{7B} (Lyu et al., 2024b)	–	33.6	56.0	73.4	–	37.6	84.4	19.1	46.4
<i>Retrieval-And-Structuring (Ours)</i>									
$\mathcal{M}_{Plan}$	$\mathcal{M}_{Ans}$								
RAS _{Sonnet-3.5}	RAS _{Sonnet-3.5}	77.4	57.7	62.3	71.4	93.8	39.1	70.5	23.3
RAS _{Sonnet-3.5}	RAS _{7B}	73.8	43.4	61.3	75.5	62.4	37.2	95.2	19.7
RAS _{7B}	RAS _{Sonnet-3.5}	78.7	54.7	58.8	70.8	91.2	39.1	70.5	23.3
RAS _{7B}	RAS _{7B}	72.6	42.1	58.3	74.8	67.7	37.2	95.2	19.7
RAS _{8B}	RAS _{8B}	73.8	44.2	57.7	77.5	71.5	37.6	96.2	20.1

Table 1. **Performance Comparison.** We highlight the top-2 closed-source models and top-2 open-source 7B models, and underline the best closed-/open-source (7B) model for each dataset. Note: Either RAS_{7B} or RAS_{8B} is a single model trained with action planner’s and answerer’s instructions via multi-task learning. No ensemble or multi-agent approach is used for any experiments.

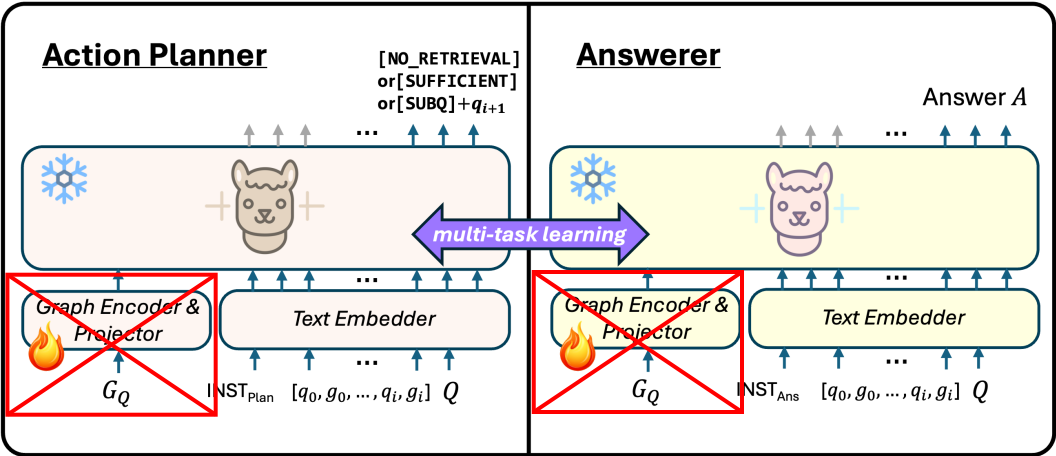
Experiments

Ablation Study

	TQA (acc)	2WQA (F1)	Pub (acc)	ASQA (rg)	ASQA (mv)
RAS _{7B}	72.6	42.1	74.8	37.2	95.2
<i>Training Phase</i>					
No GraphToken	70.2	38.4	66.4	33.1	85.0
No LoRA	71.5	37.8	54.8	32.8	84.8
No Text-to-Triple	70.4	38.2	71.4	36.2	73.8
No Mutli-Task	68.6	39.2	65.5	36.7	88.9
<i>Inference Phase</i>					
No Retrieval	56.9	27.4	69.0	31.3	70.6
No GraphToken	68.8	38.7	67.3	36.5	93.6
No Planning	66.7	37.8	71.5	–	–
No ThemeScope	72.0	–	73.9	–	–

“No GraphToken” (training) removes the graph encoder, using only LoRA-based LLM fine-tuning.

“No GraphToken” (inference) denotes empty input to the graph encoder during inference.



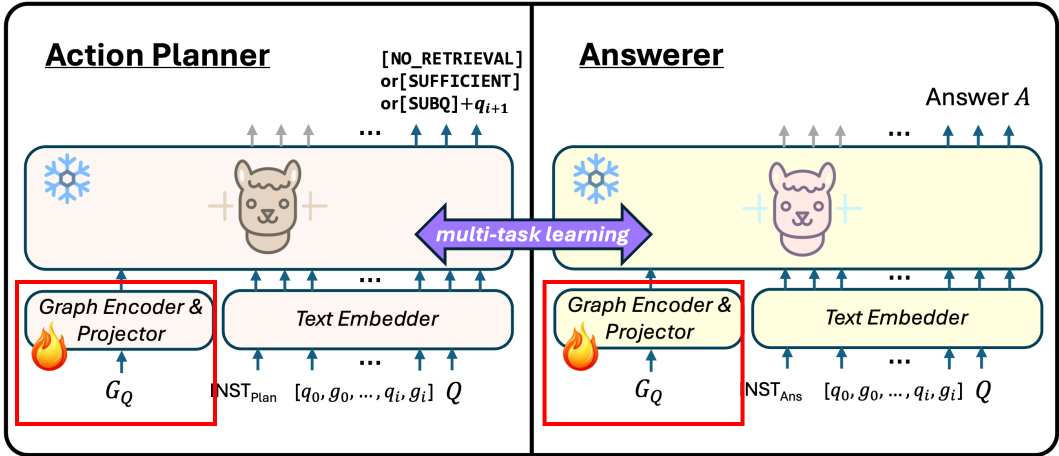
Both ablations highlight the importance of encoding the **structural information**

Experiments

Ablation Study

	TQA (acc)	2WQA (F1)	Pub (acc)	ASQA (rg)	ASQA (mv)
RAS _{7B}	72.6	42.1	74.8	37.2	95.2
<i>Training Phase</i>					
No GraphToken	70.2	38.4	66.4	33.1	85.0
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No GraphToken	68.8	38.7	67.3	36.5	93.6
No Planning	66.7	37.8	71.5	–	–
No ThemeScope	72.0	–	73.9	–	–

“No LoRA” uses graph token optimization without low-rank adaptation.



Without LoRA, it is too difficult for LLM to understand the Graph Token, i.e. to understand the encoded structure

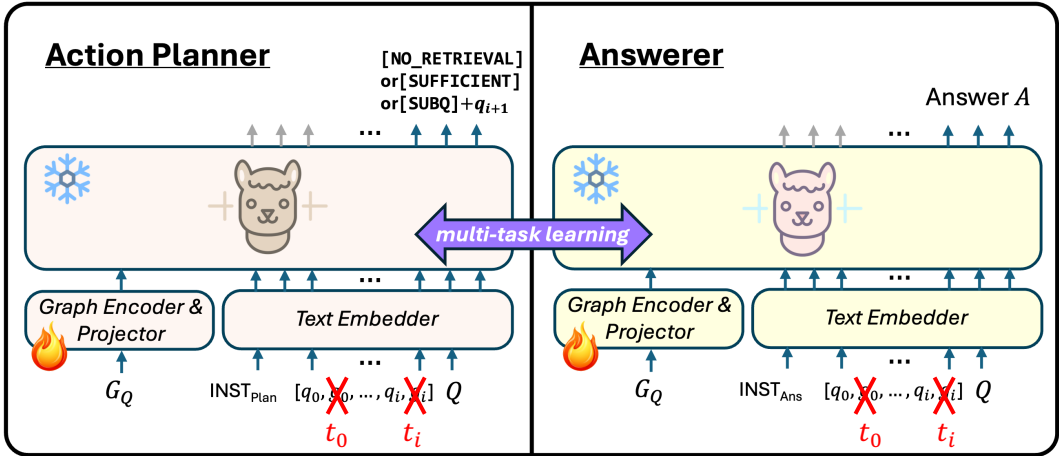
This aligns G-Retriever’s (He et al., NeurIPS 2024) finding.

Experiments

Ablation Study

	TQA (acc)	2WQA (F1)	Pub (acc)	ASQA (rg)	ASQA (mv)
RAS _{7B}	72.6	42.1	74.8	37.2	95.2
<i>Training Phase</i>					
No GraphToken	70.2	38.4	66.4	33.1	85.0
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No Planning	66.7	37.8	71.5	–	–
No ThemeScope	72.0	–	73.9	–	–

“No Text-to-Triple” keeps the original retrieved text instead of constructing KGs.



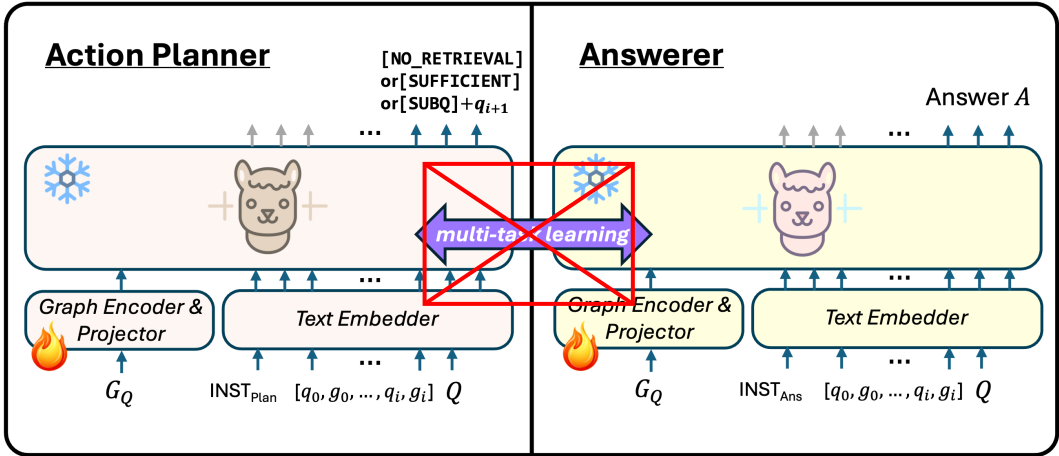
Without Triple (Structure) Extraction, the model sometimes cannot capture the most essential information in the retrieved context, which significantly impact model’s long-form generation capability.

Experiments

Ablation Study

	TQA (acc)	2WQA (F1)	Pub (acc)	ASQA (rg)	ASQA (mv)
RAS _{7B}	72.6	42.1	74.8	37.2	95.2
<i>Training Phase</i>					
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No Planning	66.7	37.8	71.5	–	–
No ThemeScope	72.0	–	73.9	–	–

No Multi-Task” trains separate action planner and answerer rather than training a single model.



Multi-task learning of action planning and answering is essential for the performance gain.

This aligns RPG’s (Lyu et al., EMNLP 2024) finding.

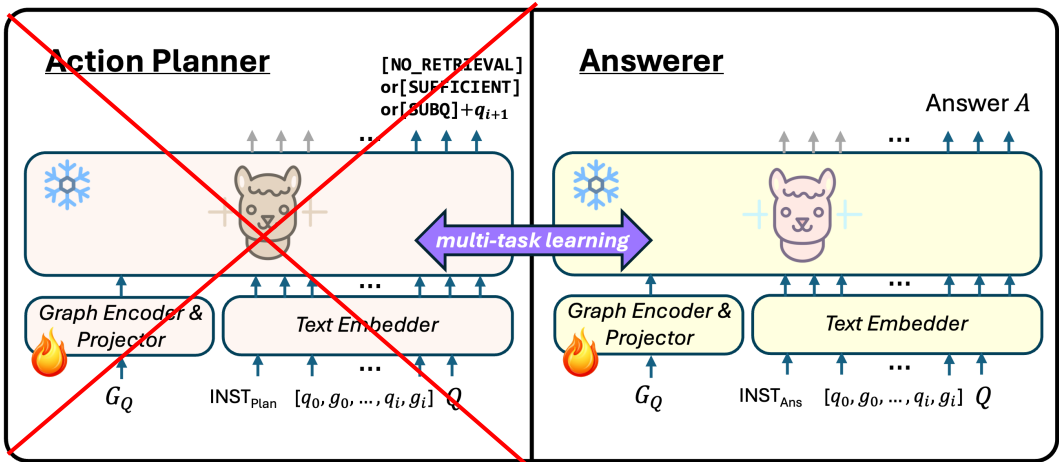
Experiments

Ablation Study

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RAS _{7B}	72.6	42.1	74.8	37.2	95.2
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No Retrieval	56.9	27.4	69.0	31.3	70.6
No GraphToken	68.8	38.7	67.3	36.5	93.6
No Planning	66.7	37.8	71.5	–	–
No ThemeScope	72.0	–	73.9	–	–

“No Retrieval” tests direct query answering without any context.

“No Planning” uses single retrieval.



It shows the importance of self-reflective retrieval.

This aligns both SelfRAG’s (Asai et al., ICLR 2024) and RPG’s (Lyu et al., EMNLP 2024) finding.

Experiments

Ablation Study

	TQA (acc)	2WQA (F1)	Pub (acc)	ASQA (rg)	ASQA (mv)
RAS _{7B}	72.6	42.1	74.8	37.2	95.2
<i>Training Phase</i>					
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No Planning	66.7	37.8	71.5	–	–
No ThemeScope	72.0	–	73.9	–	–

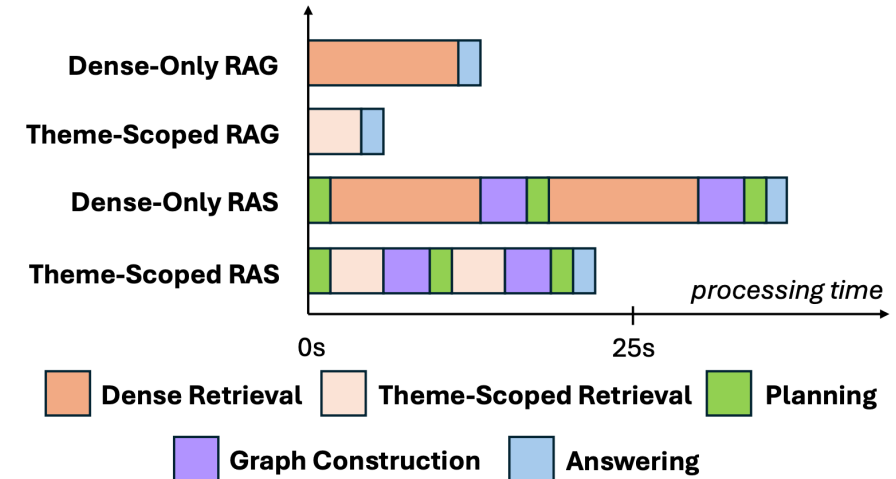


Figure 2. Efficiency study of RAS conducted on our machine. For RAS, we set #iteration=2 and use closed-source LLM for illustration. The result is averaged over 100 TriviaQA samples tested. $K=250,000$ and $k=5$ are set in this case. The Wiki 2018 dump ($\sim 32\text{M}$ documents) is used and loaded on CPU for this analysis.

Ablating Theme Scoping causes minimal performance drop.
However, Theme Scoping significantly boosts the retrieval efficiency.

Experiments

Performance Scaling

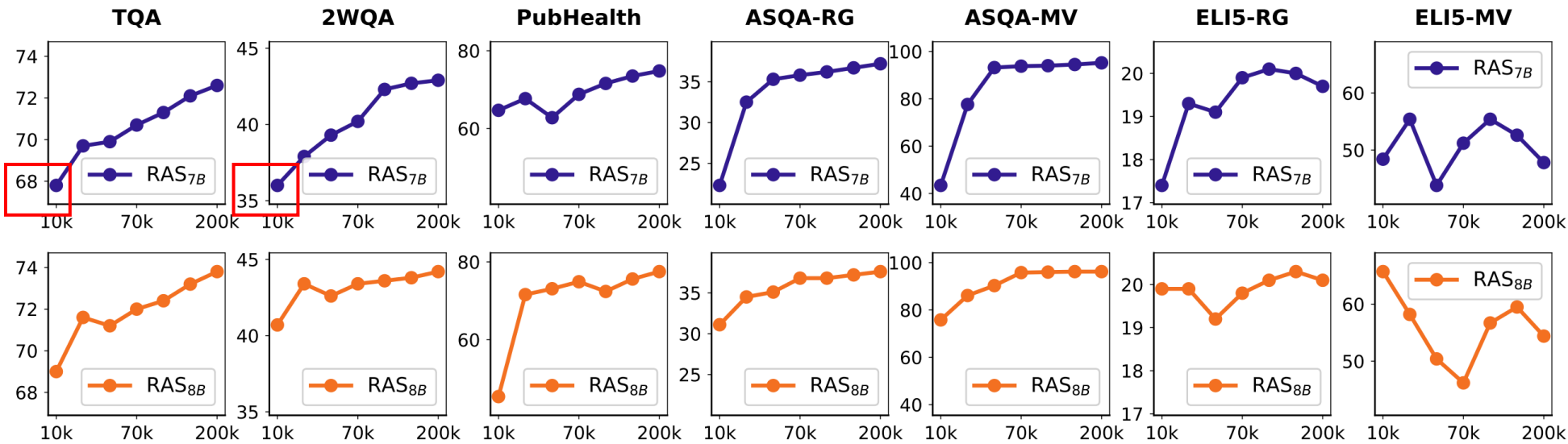


Figure 3. **Impact of Training Data Volume on Model Performance.** Results for RAS_{7B} (top) and RAS_{8B} (bottom) illustrate how performance scales with increasing training data.

	TQA (acc)	2WQA (F1)
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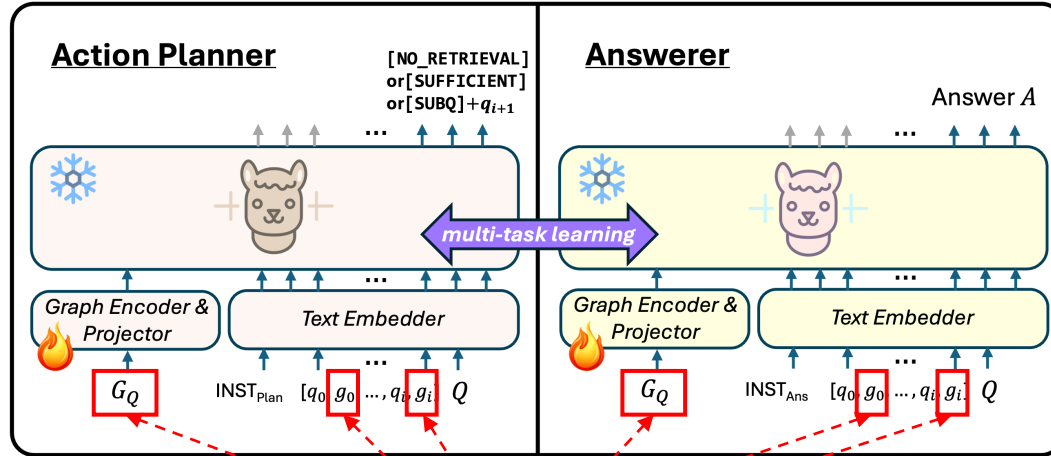
SELF-RAG_{7B} (Asai et al., 2023)
SELF-RAG_{13B}
RPG_{7B} (Lyu et al., 2024b)

66.4 25.1
69.3 –
– 33.6

- (1) Model performance generally improves with the increased training data
- (2) RAS outperforms previous methods with only 5% (10K) data on TQA and 2WQA
- (3) Performance on ELI5 doesn't scale normally (due to the inclusion of ASQA's data in the training set, following previous work)

Experiments

Graph Information Abundance vs Performance



We control the information ratio here

Further demonstrate RAS's strong capability of effectively utilizing available graph information for knowledge-intensive generation

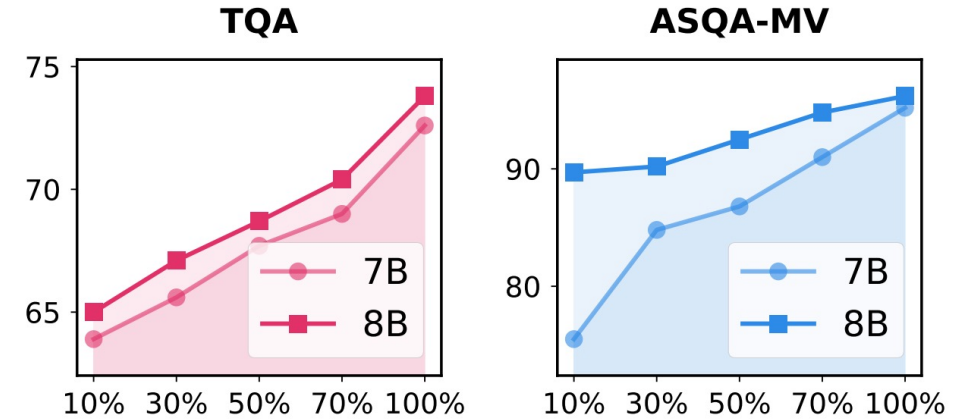
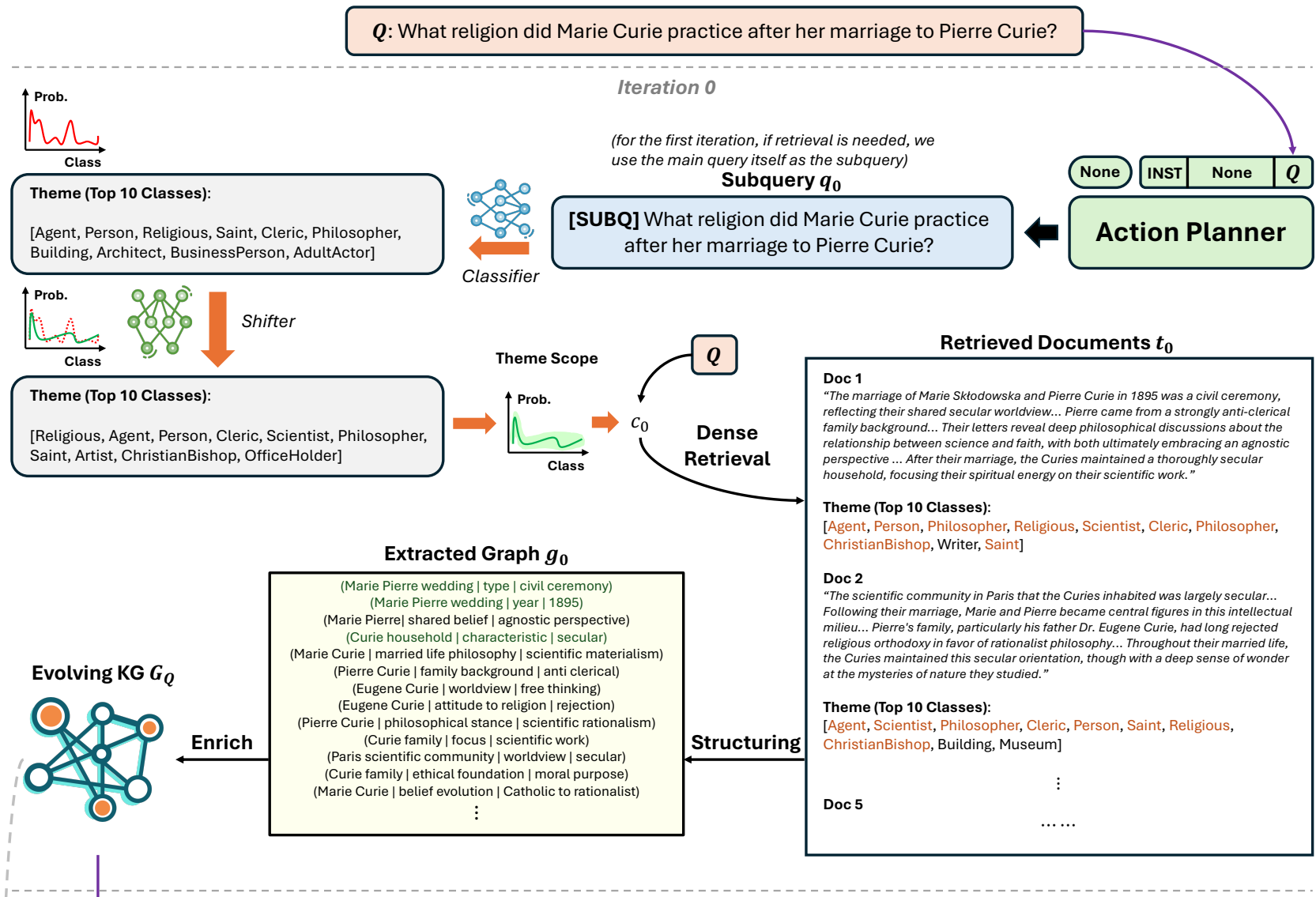
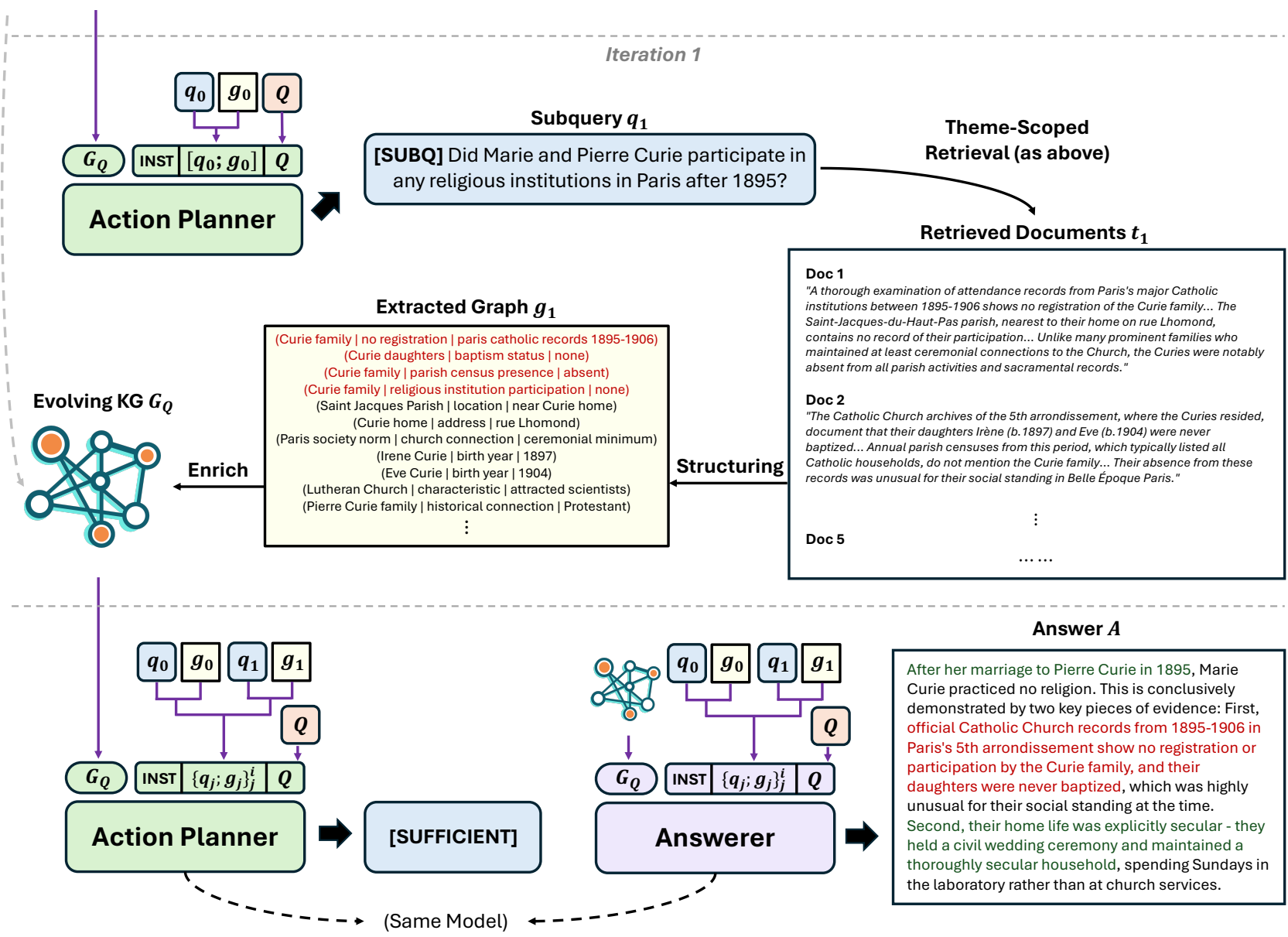


Figure 4. Impact of Graph Information Abundance on Model Performance. For each sample, we randomly shuffle its associated triples five times and take different ratios (10%, 30%, 50%, 70%, 100%) of the shuffled data. The performance scores shown are averaged across these five shuffling runs.

Experiments – Running Example



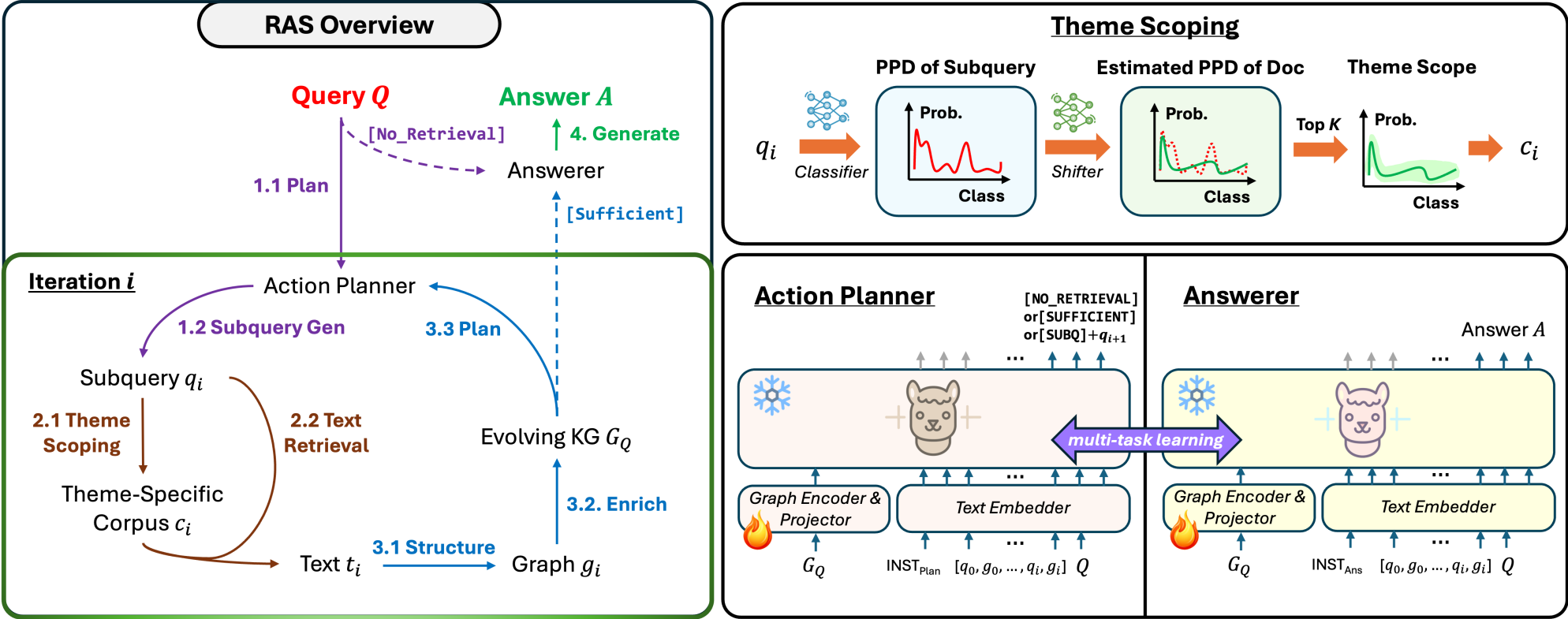
Experiments – Running Example



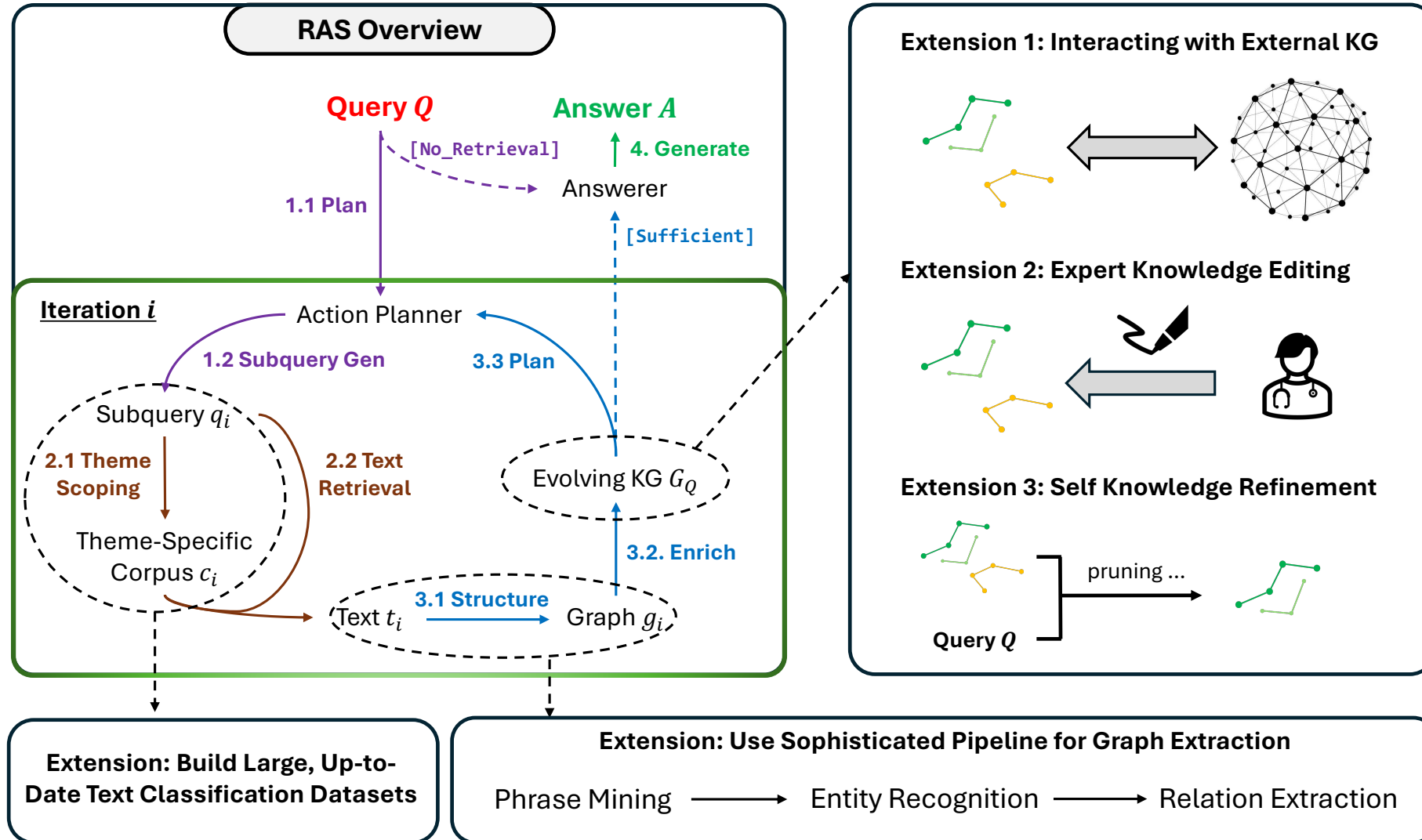
Conclusion & Future Work

Conclusion

We presented RAS (Retrieval-And-Structuring), a novel framework that enhances knowledge-intensive language generation through iterative retrieval and knowledge structuring. Our approach introduces key innovations including theme-scoped retrieval, action planning, dynamic knowledge structuring, and graph-augmented answering. Comprehensive experiments demonstrate RAS’s superior performance over existing approaches across diverse benchmarks.



Limitations & Future Work



(Problem: DBPedia-298 is old, making the theme scoping ineffective on newer corpus like wiki2020)

There are still a lot to explore!

Thank you!

<https://pat-jj.github.io/assets/pdf/ras.pdf>
<https://anonymous.4open.science/r/RAS-Anonymous>

Appendix

Data Preparation & Model Training

Training Data Example 1 (Planner – [Sufficient])

"input": "

You are a planner to determine if the question can be answered with current information and output the appropriate label as well as the subquery if needed.

Output [NO_RETRIEVAL] if the question can be directly answered with the question itself without any retrieval.

Output [SUBQ] with an subquery for retrieval if still needs a subquery.

Output [SUFFICIENT] if the question can be answered with the provided information.

[SUBQ] What gaming control board in Ohio is Martin R. Hoke a member of?

Retrieved Graph Information: ['(S> Martin R. Hoke| P> Former position| O> Member of the United States House of Representatives)', '(S> Martin R. Hoke| P> Birth date| O> May 18, 1952)', '(S> Martin R. Hoke| P> Occupation| O> Politician)', '(S> Martin R. Hoke| P> State| O> Ohio)', '(S> Martin R. Hoke| P> Nationality| O> American)', '(S> Martin R. Hoke| P> Party| O> Republican)', '(S> Martin R. Hoke| P> Member of| O> Ohio Casino Control Commission)', '(S> Martin R. Hoke| P> Born| O> 1952)']

[SUBQ] What gaming control board provides oversight of Ohio's casinos?

Retrieved Graph Information: ['(S> Ohio Casino Control Commission| P> Function| O> Provides oversight of the state's casinos)', '(S> Ohio Casino Control Commission| P> Location| O> Ohio)', '(S> Ohio Casino Control Commission| P> Type| O> Gaming control board)', '(S> Ohio Casino Control Commission| P> Abbreviation| O> OCCC)']

Question: Martin R. Hoke, is an American Republican politician, member of which gaming control board in Ohio that provides oversight of the state's casinos?

"

"label": "[SUFFICIENT]"

Data Preparation & Model Training

Training Data Example 2 (Planner – [SUBQ])

"input": "

You are a planner to determine if the question can be answered with current information and output the appropriate label as well as the subquery if needed.

Output [NO_RETRIEVAL] if the question can be directly answered with the question itself without any retrieval.

Output [SUBQ] with an subquery for retrieval if still needs a subquery.

Output [SUFFICIENT] if the question can be answered with the provided information.

[SUBQ] Where is The Pick Motor Company Limited located?

Retrieved Graph Information: ['(S> Pick Motor Company Limited| P> Alias| O> New Pick Motor Company)', '(S> Pick Motor Company Limited| P> Location| O> Stamford, Lincolnshire)', '(S> Pick Motor Company Limited| P> Operational period| O> 1899-1925)', '(S> Pick Motor Company Limited| P> Industry| O> Motor vehicle manufacturing)']

Question: The Pick Motor Company Limited is located in a town on which river ?",

"label": "[SUBQ] Which river is Stamford located on?"

Data Preparation & Model Training

Training Data Example 3 (Planner – [NO_RETRIEVAL])

"input": "

You are a planner to determine if the question can be answered with current information and output the appropriate label as well as the subquery if needed.

Output [NO_RETRIEVAL] if the question can be directly answered with the question itself without any retrieval.

Output [SUBQ] with an subquery for retrieval if still needs a subquery.

Output [SUFFICIENT] if the question can be answered with the provided information.

Question: Given a chat history separated by new lines, generates an informative, knowledgeable and engaging response. ## Input:\n\nI love pizza. While it's basically just cheese and bread you can top a pizza with vegetables, meat etc. You can even make it without cheese!\nPizza is the greatest food ever! I like the New York style.\nI do too. I like that the crust is only thick and crisp at the edge, but soft and thin in the middle so its toppings can be folded in half.\nAbsolutely! I am not that big of a fan of Chicago deep dish though.",

"label": "[NO_RETRIEVAL]"

Data Preparation & Model Training

Training Data Example 4 (Answerer – [NO_RETRIEVAL])

"input": "

You are a answerer given a question and retrieved graph information.

Each [SUBQ] is a subquery we generated through reasoning for the question. The retrieved graph information follows each [SUBQ] is relevant graph information we retrieved to answer the subquery.

[NO_RETRIEVAL] means the question can be answered with the question itself without any retrieval.

The main question starts with \"Question: \". Please answer the question, with subqueries and retrieved graph information if they are helpful.

[NO_RETRIEVAL]

Question: Which person won the Nobel Prize in Literature in 1961, Ivo Andri or Nicholas Pileggi?",

"label": "Ivo Andri"

Data Preparation & Model Training

Training Data Example 5 (Answerer – [SUFFICIENT])

"input": "

You are a answerer given a question and retrieved graph information.

Each [SUBQ] is a subquery we generated through reasoning for the question. The retrieved graph information follows each [SUBQ] is relevant graph information we retrieved to answer the subquery.

[NO_RETRIEVAL] means the question can be answered with the question itself without any retrieval.

The main question starts with \"Question: \". Please answer the question, with subqueries and retrieved graph information if they are helpful.

[SUBQ] What type of athlete is Darold Williamson?

Retrieved Graph Information: ['(S> Darold Williamson| P> Nationality| O> American)', '(S> Darold Williamson| P> Birth date| O> February 19, 1983)', '(S> Darold Williamson| P> Occupation| O> Track athlete)']

[SUBQ] What specific skills are included in track and field events?

Retrieved Graph Information: ['(S> Athletics| P> Includes| O> Road running)', '(S> Track and field| P> Includes| O> Throwing)', '(S> Track and field| P> Includes| O> Jumping)', '(S> Athletics| P> Includes| O> Cross country running)', '(S> Track and field| P> Categorised under| O> Athletics)', '(S> Track and field| P> Includes| O> Running)', '(S> Track and field| P> Venue| O> Stadium with an oval running track enclosing a grass field)', '(S> Athletics| P> Includes| O> Track and field)', '(S> Athletics| P> Includes| O> Race walking)', '(S> Track and field| P> Based on skills of| O> Running)', '(S> Track and field| P> Based on skills of| O> Jumping)', '(S> Track and field| P> Based on skills of| O> Throwing)']

Question: Darold Williamson is an athlete in what running and jumping sport?",

"label": "Track and field"